

THE PERMISSION PARADOX

Longview Advisor: An AI Agent for
Reviewing the Future Potential of New Ideas



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Future Potential of New Ideas*

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Author's Note

This work was created under the umbrella of Axiologic Research as part of two interconnected research programs, one dedicated to Artificial Intelligence and the other to Outfinitism, both led by Sinică Alboaie. To explore the details of how these two programs intersect and build upon each other, we invite readers to visit the Outfinity Venture Studio page at www.outfinity.ch.

While Artificial Intelligence language models were used to assist with text generation, the foundational philosophy, thematic structure, and analytical approach derive exclusively from the author's decades of dedicated study of social phenomena, social technologies, and genuine hard technologies resulting from various European and self funded research projects. Recently, within the Achilles research project, we have been deeply studying AI generated literature and its automated verification using neuro symbolic approaches; all the free books made available to the public are part of the suite of works we use to test our latest technologies.

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A Note on Evidence, Values, and Technology Status

This book begins with a grievance that is recognizable and easy to romanticize. People working at the edge of a field often feel that their ideas must ask permission from reviewers, investors, regulators, customers, and senior experts who cannot yet see what the ideas could become. Sometimes the grievance is justified; sometimes the innovator is wrong. More often, the conflict reflects a structural mismatch: the innovator reasons backward from a possible future, while the evaluator reasons forward from evidence produced by the past.

The argument does not divide society into visionaries and fools. Human beings depend on social learning, reputation, precedent, and institutions because no person can verify more than a small fraction of modern knowledge. Copying helps create cumulative culture, and expertise compresses hard-won experience into usable judgment. Yet every compression loses information, and every institution trained on yesterday develops blind spots about tomorrow.

Artificial intelligence sharpens the problem by lowering the cost of hypotheses, software, scientific prose, designs, simulations, and convincing explanations. Candidate discoveries may soon arrive faster than laboratories, reviewers, investors, regulators, and communities can test or absorb them. The Discovery Flood is therefore a scenario about rate mismatch, not a claim that every AI-generated idea is valuable or that endless breakthroughs are guaranteed.

The normative position is equally explicit. Institutions must protect people, allocate scarce resources, demand evidence, and prevent irreversible harm. But their authority is stewardship, not

ownership of unexplored possibility. When potential benefit and learning value are high, downside is bounded, and an experiment is reversible, preventing the test has an opportunity cost and requires justification.

The response proposed here combines better implication analysis with a different funding grammar: idea-first review, smaller and shorter commitments when the epistemic clock permits, patient renewal while difficult work continues to produce information, qualified lotteries where reviewers cannot discriminate reliably, visible failures, and continuous evidence review instead of ceremonial reporting at fixed dates.

The standard is not that every strange idea deserves belief. It is that consequential possibilities deserve something better than premature compression into familiar categories. The future should not receive a free pass. It should receive a fair test.

Longview Advisor is a proposed technology and research programme, not a finished product and not an autonomous funding authority. The book treats its LLM-heavy and neuro-symbolic architectures, its integration with Executable Science, and its institutional use cases as falsifiable engineering hypotheses. The goal is to make prospective judgment more inspectable and useful to human reviewers, not to move discretionary power into an unchallengeable model.

CHAPTER 1

The Book in One Chapter

1.1 The Discovery Flood and the Permission Paradox

The central danger of the next technological era may not be a shortage of ideas. It may be that discovery becomes abundant before understanding does.

For most of history, a coherent proposal was costly to produce. A scientific hypothesis required years of training, a literature review took weeks, a prototype required a team, and polished prose signaled at least some investment of thought. AI is dissolving these correlations. Human-AI teams can already inspect large literatures, generate alternatives, implement software, design experiments, and present complete-seeming strategies at speeds that recently required an institution. The discontinuity is not merely faster assistance. It is a change in the ratio between candidate novelty and society's capacity to examine it.

This is the Discovery Flood: a potentially continuous, distributed, and cumulative inflow of hypotheses, molecules, architectures, instruments, business models, policy mechanisms, and combinations of existing knowledge. Much of the flow will be trivial, repetitive, wrong, or dangerous. Some of it may be extraordinarily valuable. The institutional problem is that generation can scale faster than discriminating attention, physical validation, replication, regulation, and adoption.

The permission paradox predates AI. The farther an idea lies from the present, the fewer of the present's credibility signals it can possess. It lacks users because the product does not exist, citations because the field has not formed, benchmarks because the

phenomenon is not yet measured, and prestigious sponsors because prestige was earned in the old paradigm. The idea needs resources to create evidence but needs evidence to receive resources. AI adds a second paradox: it makes unfamiliar ideas easier to describe and harder to distinguish. When every proposal can sound complete, reviewers may rely even more on reputation, social proof, inherited categories, and standardized scores.

The mismatch has five dimensions. Generation accelerates while validation remains slow. Fluent summaries create semantic closure, the feeling of understanding after decisive implications have been compressed away. Institutions allocate scarce attention through reputational proxies. Large, long grants reward confident prediction where frontier work needs staged learning. Finally, delay is treated as neutral even when it selects concentration, stagnation, or a worse technological path.

None of this makes the innovator presumptively right. Strange ideas can be confused, unsafe, uneconomic, or merely strange. The answer is not admiration but higher-resolution review. Instead of asking only whether a document deserves belief, institutions should ask what would have to be true for the idea to matter, which worlds follow under those conditions, and what bounded experiment could most efficiently change the decision.

The proposed technological response is the Longview Advisor: a versioned second-order review workflow combining language models, evidence retrieval, causal and symbolic representations, executable tests, value lenses, and human judgment. It reconstructs the proposal faithfully, preserves alternative interpretations, maps consequences and stakeholders, exposes assumptions, and converts uncertainty into tests. It works before funding, during active research, and after results, so review becomes a living argument connected to evidence rather than a sequence of disconnected documents.

The institutional response is permissionless proof and permissioned scaling. Bounded, reversible, instrumented inquiry should be easier to authorize; evidentiary demands should rise with scale, irreversibility, and exposure of non-consenting parties. Grants should often become smaller, faster, and renewable. Review should follow epistemic events rather than calendar ceremonies. Failures should remain visible when they eliminate hypotheses. Reputation should be separated from the evaluation of the idea wherever possible.

The urgency is political as well as scientific. Public institutions that respond to acceleration mainly by slowing decisions will not stop experimentation. They will cede it to actors that already control capital, data, compute, laboratories, distribution, and legal capacity. A civilization can miss its future politely: commissioning reports, celebrating innovation, and rejecting every proposal that lacks precedent. The alternative is not to worship novelty. It is to build institutions capable of hearing it.

1.2 The Proposed Response: Longview, Reversible Inquiry, and a New Funding Grammar

Longview therefore sits between two institutional failures. One is premature rejection: the strange idea is compressed into a familiar category, scored against inappropriate comparators, and removed before reality can test it. The other is technological credulity: a persuasive AI system invents an exciting future around a weak proposal. The design goal is not optimism. It is disciplined conditionality. The system should be able to say that a proposal is poorly evidenced and still worth a small experiment, or that a fashionable proposal has strong rhetoric but little option value.

The book's institutional proposal follows the same logic. AI should make research bureaucracy shrink as fast as scientific possibility expands. Where AI reduces the cost of literature review, coding,

simulation, documentation, and iteration, many grants can become smaller, shorter, and staged around evidence. Yet difficult research must retain the possibility of long patience. Real science must be able to fail, and productive failure can justify continuation when it eliminates hypotheses, reveals constraints, or creates reusable tools. Short cycles and long patience are complements, not opposites.

The first distinctive claim is that the conflict is not best understood as visionaries against conformists. Innovators reason from mechanisms and possible futures; evaluators reason from base rates, precedents, and evidence they can defend to others. Both forms of rationality are necessary. The permission paradox appears when institutions have no reliable way to convert one into the other except premature judgment.

The second claim is the novelty-absorption gap. AI can generate and articulate candidate novelty at machine speed, while empirical validation, safety assurance, physical experimentation, replication, institutional learning, and cultural adoption retain human and material clocks. The resulting verification lag is not merely a delay between discovery and use. It is a widening gap between the number of plausible claims society can produce and the number it can responsibly test.

The third claim combines semantic closure with second-order blindness. Fluent AI summaries may reduce the felt need for further inquiry while translating a discontinuous idea into the nearest familiar category. Ordinary review can then be locally correct and strategically blind: it asks whether the first-order claim is supported but not what new actions, institutions, markets, scientific paths, or risks become possible if it succeeds. The Longview Advisor is designed as an anti-summary: a navigable record that keeps source claims, inferences, evidence, scenarios, values, disagreements, and decisive tests separate.

The fourth claim concerns institutional authority. Gatekeepers have duties, but their authority is contingent and delegated. The status quo is not a zero-risk option, and omission has consequences. The duty of reversible inquiry is deliberately limited: where plausible benefit and information value are high, harms are bounded, and the test can be stopped, institutions should need a reason to block inquiry rather than merely a reason to permit it.

The fifth claim is that funding should follow the epistemic clock. Many AI-accelerated projects can begin with short evidence tranches rather than multi-year predictions. Difficult programs may nevertheless require long patience through repeated renewals. Qualified lotteries can replace false precision when proposals pass a quality threshold but reviewers cannot rank them reliably. Reputation firewalls can assess the idea before introducing institutional pedigree, while validation commons fund replication, benchmarks, provenance, negative results, and executable research infrastructure.

The sixth claim addresses machine-discovered knowledge. Some valid results may be difficult to compress into human concepts. The relevant distinction is not merely interpretable versus opaque. Human-legible science supports conceptual understanding; machine-legible science may resist intuition but remains executable, reproducible, inspectable, and challengeable; machine-opaque claims provide neither a sufficient explanation nor an independent verification path. Scientific institutions should pursue human understanding without making human comfort the sole criterion of truth.

The practical conclusion is asymmetric. Novel ideas do not deserve automatic acceptance, but bounded possibilities deserve more than automatic compression. Innovators should anchor novelty, demonstrate before declaring, build dense early communities, and match capital to uncertainty. Institutions should route uncertain

proposals toward the cheapest decisive evidence instead of converting every ambiguity into rejection. Under a Discovery Flood, the central measure of progress is not ideas per hour but decisive tests per unit of unexplored possibility.

CHAPTER 2

Why a Species That Copies Can Still Invent

2.1 Social Learning, Prestige, and the Wisdom of Deference

Human beings do not become capable by thinking independently about everything. They become capable by inheriting an immense body of knowledge that no individual understands in full. Language, agriculture, mathematics, law, medicine, computing, and scientific practice are cumulative achievements. Each depends on people following procedures, trusting sources, and reproducing solutions whose causal foundations they may only partially grasp.

Research on cultural evolution therefore treats social learning as a central human adaptation. Boyd, Richerson, Henrich, and others have argued that our ecological success cannot be explained by individual intelligence alone. A person placed outside a cultural tradition cannot simply reason a modern society back into existence. Knowledge is distributed across minds, artifacts, institutions, and routines. Copying is the mechanism by which solutions survive the death of their inventors and become platforms for further invention.

This makes deference rational. The world is too complex for direct verification. A patient cannot reproduce every clinical trial. An engineer cannot inspect every layer of a global software stack. An investor cannot master every scientific discipline represented in a portfolio. People therefore use proxies. They look at expertise, reputation, institutional affiliation, consensus, adoption, and the behavior of trusted peers. Under ordinary conditions, these shortcuts save time and often improve decisions.

The empirical literature does not support a caricature in which most people imitate blindly. People combine personal and social information, often increasing reliance on others when their own evidence is weak. The strategy sometimes described as "copy when uncertain" is sensible. When direct observation is expensive or noisy, the behavior of others contains information. Individual differences, incentives, domain knowledge, and perceived competence all shape how much social information is used.

Prestige is one especially efficient signal. If many people voluntarily attend to a person, defer to her, or seek access to her knowledge, a newcomer can infer that she may possess valuable competence. Prestige-biased learning compresses a complicated evaluation into a socially observable cue. Yet the cue is retrospective. Prestige is earned through performance in an environment that already existed. During a discontinuity, the person best adapted to the old regime may not be the person best able to see the new one.

This does not make experts enemies of progress. It means that expert performance and frontier anticipation are different variables. A senior scientist may be the best evaluator of whether a proposed mechanism violates established evidence and a poor evaluator of whether a new research instrument will create an unfamiliar field. A procurement officer may understand organizational constraints better than the inventor and still underestimate the possibility that the constraints themselves will change. The proper response is not to discard expertise but to compose different forms of it.

René Girard's theory of mimetic desire offers a more ambitious claim: people do not merely copy actions; they often learn what to want through models and rivals. Contemporary experiments support narrower versions of this intuition. Social information can shift private preferences, and people tend to move toward the tastes of valued groups while distancing themselves from disliked

groups. Yet it would be unjustified to conclude that human desire is nothing but imitation. Biological needs, personal experience, temperament, reflection, and local circumstances matter. Mimetic theory is most useful here as a warning that social proof can alter valuation itself, not only public behavior.

The same mechanism appears in innovation. Once an idea acquires visible sponsors, it becomes easier for others to see its value. Funding does not merely provide resources; it provides a model of desire. A respected laboratory, investor, or institution signals that the proposal is worth attention. The signal can be deserved, but it can also become self-reinforcing. Projects that receive early legitimacy attract talent, citations, customers, and more capital. Projects that lack it may remain invisible even when their underlying mechanism is strong.

The compounding effect is measurable. Bol, de Vaan, and van de Rijt compared early-career researchers who fell just above and just below a funding threshold with nearly identical review scores. Winners accumulated more than twice as much funding during the following eight years. The advantage was not fully explained by achievements produced by the initial grant. Early recognition itself became an asset, while rejection also reduced later participation. A narrow decision therefore helped create the future reputation that subsequent decisions treated as evidence.

At the same time, fine-grained review rankings often imply more certainty than the process can support. In an experiment in which different groups assessed the same NIH applications, Pier and colleagues found no agreement about application quality in either qualitative or quantitative judgments. This does not prove that peer review is useless. It shows that close rankings can contain substantial reviewer noise, especially when proposals are genuinely difficult to compare.

Reputation can be separated from the idea at least temporarily. When the Arnold and Mabel Beckman Foundation blinded names and institutions during the first stage of a young-investigator competition, Hultgren and colleagues found a reduction in institutional-prestige bias that continued into later stages. The result does not imply that execution capability is irrelevant. It suggests a better sequence: first ask whether the possible future deserves investigation; only then ask whether this team and environment can execute the test.

This sequence becomes more important under AI. As proposals become uniformly polished, language will reveal less about the underlying work. Evaluators may compensate by leaning more heavily on affiliation, prior grants, famous collaborators, and endorsements. AI can democratize production while aristocratizing selection. A deliberate reputation firewall is therefore not anti-expertise. It is a way to keep inherited status from answering a question that should initially be answered by the mechanism and the proposed evidence.

This is not a defect that can be removed by exhorting people to think for themselves. It is a coordination structure. Civilization needs reliable ways to decide whose claims deserve costly attention. The question is whether the structure preserves enough routes by which an idea can earn legitimacy without already possessing it.

The exploration-exploitation trade-off captures the larger balance. Exploration searches unfamiliar regions and tolerates failure. Exploitation scales solutions already known to work. A population composed entirely of explorers would fail to stabilize useful practices. A population composed entirely of imitators would converge efficiently and stop discovering. Cultural progress depends on a division of cognitive labour in which a minority

generates variation, a larger population evaluates and adopts, and institutions preserve successful solutions.

The problem begins when the division hardens into hierarchy. Explorers become supplicants, evaluators become sovereigns, and procedures designed to filter noise acquire the power to define what counts as signal. Under stable conditions, this may remain tolerable. Under rapid technological change, the cost of the arrangement rises because the evidence used by evaluators becomes stale faster.

AI intensifies this dynamic in two directions. It gives more people exploratory capacity by reducing the cost of generating and testing ideas. At the same time, it increases the amount of material that evaluators must filter. The predictable response is greater use of proxies: famous institutions, familiar framings, benchmark scores, model-generated summaries, and the visible decisions of other funders. The species that copies gains a machine for variation and may answer by copying more aggressively.

The lesson is not that imitation must be defeated. Innovation needs imitation to spread. The institutional task is to delay convergence until enough alternatives have encountered reality. A healthy society does not ask everyone to become a visionary. It gives visionaries enough room to be wrong in informative ways.

2.2 Cascades, Premature Convergence, and Institutional Blindness

A network can become less intelligent by learning too quickly.

The intuition that faster communication improves collective problem solving is often correct. Information spreads, duplication falls, and useful discoveries become available to more people. Yet experiments on cultural accumulation show a countervailing effect. In fully connected groups, participants can observe successful

solutions rapidly and converge on them. The convergence looks efficient, but it removes the independent search paths from which a superior solution might have emerged. Partially connected groups can preserve local variation long enough to discover more complex outcomes.

This pattern is closely related to information cascades. A person observes the choices of others and rationally infers that those choices contain private information. After enough people choose the same option, later participants may ignore their own weaker signals and follow the visible majority. The cascade can aggregate knowledge, but it can also lock a group into error. Once behavior is public and reasoning is hidden, repetition becomes difficult to distinguish from independent confirmation.

Modern institutions are cascade machines. Citation counts, rankings, venture rounds, procurement frameworks, journal prestige, standards, and recommendation algorithms turn prior choices into visible evidence. Each signal helps people navigate abundance. Together they can cause a small initial advantage to become an epistemic monopoly. The institution does not need to prohibit alternatives. It merely needs to allocate attention in proportion to attention already received.

Uncertainty makes the effect stronger. Experiments by Mueller, Melwani, and Goncalo found that people can express a desire for creativity while exhibiting an implicit bias against creative ideas when uncertainty is salient. Status quo bias gives the existing option an additional advantage: it is not evaluated as one proposal among equals but as the default from which change must justify itself. New ideas therefore compete on an uneven field. The old system arrives with infrastructure, vocabulary, training, social proof, and the absence of a visible transition cost. The new system arrives carrying its own risks and is often blamed for the costs of leaving the old one.

Institutions magnify these asymmetries because their decisions must be defensible. A reviewer can justify choosing an established method by citing standards, precedents, and expert consensus. Supporting an unfamiliar proposal requires a longer explanation and exposes the decision-maker to asymmetric blame. If the conventional project fails, the failure is ordinary. If the strange project fails, the evaluator may appear reckless. This is career risk, not necessarily intellectual incapacity.

The permission paradox emerges from this asymmetry. A radical proposal lacks the social and institutional signals used to decide which proposals deserve a closer look. It cannot point to a market that would exist only after adoption, a benchmark that must be invented, a mature field of experts, or a long record of safe deployment. The proposal may need to change the evaluator's categories before it can be evaluated within them.

At this point innovators often make a strategic error. They interpret resistance as proof of their own depth and explain the vision at greater abstraction. The evaluator becomes less able to connect the proposal to observable evidence, and both sides leave confirmed in their view of the other. The innovator sees timidity. The evaluator sees grandiosity.

A better interpretation is that unfamiliar ideas pay several taxes simultaneously. They pay an epistemic tax because evidence is incomplete, a legibility tax because the category is unstable, a legitimacy tax because few respected actors endorse it, a coordination tax because surrounding systems must change, a reversibility tax because failure may be costly, and an identity tax because adoption can threaten the status of people fluent in the old regime. Technical superiority on one dimension does not automatically pay these debts.

The answer is not to make the idea ordinary. It is to reduce the taxes sequentially. A demonstration can reduce epistemic

uncertainty. A familiar interface can reduce legibility cost. A respected early user can transfer legitimacy. A bounded pilot can reduce irreversibility. An integration layer can reduce coordination cost. A narrative that treats incumbents as contributors to the transition rather than as obsolete obstacles can reduce identity threat.

There is also an institutional answer: preserve partial independence. Grant panels can include evaluators from adjacent paradigms rather than a single disciplinary center. Funding programs can reserve a portion of the portfolio for projects selected through different criteria. Organizations can ask teams to explore independently before sharing solutions. Review systems can separate assessment of internal validity from assessment of possible enabling value. Rejected proposals can be revisited when enabling technologies change.

The goal is not maximal diversity forever. Eventually, societies must coordinate, standardize, and scale. The goal is to prevent social selection from running faster than empirical selection. Reality should eliminate alternatives after they have been tested, not before they have become legible.

This becomes urgent under AI because dense informational networks no longer connect only people. Models train on overlapping corpora, agents retrieve the same highly ranked documents, reviewers use related systems, and synthesized conclusions are fed back into future synthesis. Apparent independent agreement may be correlated reproduction. A thousand agents can form one epistemic lineage.

AI can also protect variation if designed differently. Independent models can be given different evidence sets and objectives. Search can deliberately include neglected literature and dissenting mechanisms. Evaluation can preserve disagreement rather than average it away. Experimental portfolios can be generated to

maximize information gain across competing hypotheses. But these outcomes will not appear automatically. The easiest commercial path is smooth convergence: one answer, one score, one confident recommendation.

A civilization that wants discovery must resist that convenience. Its institutions should be judged partly by how long they can keep promising alternatives alive without losing the capacity to decide. This is not indecision. It is disciplined non-convergence.

CHAPTER 3

The Discovery Flood

3.1 When Novelty Outruns Absorption

The phrase "artificial intelligence for science" once referred mainly to specialized models that predicted structures, classified images, or accelerated simulations. The frontier is moving toward systems that participate in the organization of inquiry itself. They search literatures, formulate hypotheses, compare alternatives, write experimental code, design protocols, analyze data, and decide what to try next. The change is not the arrival of an infallible machine scientist. It is the conversion of several formerly scarce cognitive operations into scalable services.

Google's AI co-scientist illustrates the direction. Its architecture uses specialized roles for generation, reflection, ranking, evolution, proximity analysis, and meta-review, coordinated around a research objective. The system generates candidate hypotheses, subjects them to iterative criticism and competition, and can incorporate tools and expert feedback. Google's Empirical Research Assistance system addresses a different bottleneck by generating and optimizing executable scientific software against empirical criteria. FutureHouse's Robin workflow coordinates literature and analysis agents with preclinical validation. ARIA is funding projects that connect AI scientists to automated laboratories and describes the resulting transition in unusually direct terms: hypothesis generation is becoming abundant, while the capacity to test ideas in the physical world is becoming the bottleneck.

These systems remain early and uneven. Laboratory validation is domain-specific, expensive, and often difficult to interpret. An AI-generated hypothesis can be novel in wording while trivial in mechanism. Automated ranking can correlate with the preferences

of the models that produced the candidates. Expert evaluation is vulnerable to the same legibility pressures discussed throughout this book. A successful demonstration in biomedicine does not imply general scientific autonomy. Yet institutions do not need certainty about machine science to confront the asymmetry already forming: the cost of generating research options is falling much faster than the cost of establishing which options deserve belief.

The immediate consequence is a novelty-absorption gap. In software-dominant fields, some candidate ideas can be compiled, tested, benchmarked, and rejected quickly. In chemistry, materials science, medicine, ecology, and engineering, reality has a slower clock. Instruments, samples, approvals, manufacturing, longitudinal outcomes, and replications cannot be generated merely by increasing inference compute. Even where experiments are computational, a benchmark can be optimized without representing the underlying scientific objective. The Discovery Flood therefore does not consist of discoveries in the strict sense. It consists of an expanding mixture of conjectures, partial results, designs, arguments, prototypes, and real breakthroughs competing for a validation capacity that does not scale at the same rate.

Publication is not a solution to this queue. AI can also accelerate writing, formatting, literature synthesis, and review. If the number of polished papers grows faster than the number of careful replications, the visible output of science can increase while the ratio of tested knowledge to articulated possibility falls. This does not mean that more papers are intrinsically bad. It means that prose is no longer a reliable approximation of epistemic effort.

The same transformation applies beyond science. Entrepreneurs can generate complete product strategies, technical architectures, customer personas, policy analyses, and investor materials before conducting a serious encounter with a user. Regulators can commission rapid syntheses of unfamiliar technologies. Investors

can evaluate a market through machine-produced research. The apparent completeness of the artifacts conceals an important asymmetry: producing a plausible world model is easier than discovering whether the world will cooperate.

The flood metaphor is more useful than the image of an AI assistant because it describes a change in the environment, not merely a faster worker inside the old workflow. Many sources can feed the flow continuously; the level can rise gradually until routines that once seemed adequate become bottlenecks; and the institutional failure comes from a mismatch between inflow and absorption capacity. The metaphor also preserves the positive side of the phenomenon. Water can destroy, but it can also irrigate, connect, and create new productive terrain. The objective is not to stop the flow. It is to build channels through which valuable novelty can move, reservoirs in which uncertain ideas can be tested without demanding immediate adoption, and measurement systems that reveal where attention is most valuable.

One response is to automate review at the same scale. This is necessary but insufficient. A language model can check consistency, search for related work, detect missing citations, estimate feasibility, and produce a critique. Yet automated evaluation can reproduce the biases of the generated material. Research on LLM judges has documented sensitivity to answer order, verbosity, model identity, and other presentation features. Work on self-correction shows that a model's confidence and iterative reflection do not provide external truth. Multiple agents that share training data and objectives can produce the appearance of debate while preserving correlated blind spots.

The solution is not simply more reviewers or faster summaries. It is a different architecture of validation. Candidate generation should be coupled to evidence retrieval, executable tests, independent measurements, provenance, model diversity, and explicit records of

uncertainty. Review should produce experiments, not only scores. A high-impact but uncertain proposal should not be flattened into an average rating; it should be routed toward the observation most likely to change the decision. In flood terms, triage should become routing rather than disposal.

This suggests a shift from document review to claim logistics. A scientific proposal contains claims of different kinds: descriptive claims about existing evidence, causal claims about mechanisms, engineering claims about feasibility, economic claims about adoption, and value claims about what outcomes matter. Each requires a different validation path. The document is a container. The epistemic work begins when the claims are separated and connected to the evidence, code, protocols, assumptions, and stakeholders that could support or defeat them.

Executable science is one name for an infrastructure built around this shift. It does not imply that all science can be reduced to software or formal proof. It means that as much of the research object as possible becomes inspectable and actionable by machines: claims linked to source spans, data linked to transformations, methods linked to code and protocols, conclusions linked to uncertainty, and proposed impact linked to explicit value assumptions. The objective is not a paper that can be read by a computer. It is a research process in which computers can help locate the exact boundary between what has been shown and what is merely being said.

Such infrastructure becomes more valuable as generation accelerates. Without it, institutions will protect themselves from abundance through cruder triage. They will reject more proposals unread, rely more heavily on brands, and automate judgments through models trained on accepted work. The result can be a perverse selection loop: AI expands the frontier of novelty while AI-mediated evaluation pulls attention back toward the semantic

center of the past. The Discovery Flood would then increase the number of possibilities produced while decreasing the fraction that receive serious independent examination.

That would be an extraordinary historical failure: building machines capable of enlarging the search space of science and technology, then connecting them to institutions optimized mainly to reduce the queue.

The strategic objective is therefore not to reduce the inflow until current institutions feel comfortable. It is to expand epistemic absorption capacity. More laboratories must become automation-compatible. More computational experiments must be reproducible. More public-good datasets, benchmarks, and standards must be created. More reviewers must be able to trace claims to evidence and implications. More funders must support infrastructure whose value appears across a field rather than inside one product. More proposals must be routed into bounded experiments rather than forced into premature yes-or-no judgments. The rate of responsible experimentation must rise with the rate of plausible conjecture.

The ratio that matters is not ideas per hour. It is decisive tests per unit of unexplored possibility.

3.2 Semantic Closure and the Human-Legibility Limit

A traditional summary compresses detail so that a reader can retain structure and decide where deeper attention is needed. AI can perform this function extraordinarily well. It can also make the compression feel so complete that the reader no longer experiences the absence of what was discarded.

This is semantic closure. The answer has the right vocabulary, balanced considerations, and a confident conclusion. It appears to

have metabolized the subject. The reader's uncertainty falls, but the world has not become less uncertain; only the reader's psychological state has changed. Generative AI adds an articulate external explanation to the familiar illusion that access to an explanation is equivalent to understanding.

Early evidence supports a precise concern rather than a universal indictment. Studies of AI-assisted knowledge work suggest that confidence in generative systems can reduce reported critical engagement and move human effort toward verification and integration. Research on scientific summarization finds overgeneralization beyond the source's supported conditions. Other work warns that AI can produce illusions of understanding and methodological monocultures even while increasing output. The common mechanism is a sharp fall in the cost of apparent comprehension.

Radical proposals are unusually vulnerable because their value often lies in relations that are not yet conventional: an unusual combination of fields, a missing institution, a changed unit of analysis, or an implication that appears only after several steps. A helpful general-purpose model may translate the proposal into the nearest familiar category. Conceptual smoothing need not insert an obvious falsehood. It can preserve the words while rounding off precisely the incompatibility that made the idea important.

The danger increases when AI mediates every transition from source to decision. A book becomes a memo, the memo an investment note, the note a dashboard, and the dashboard a recommendation. Each compression can be locally reasonable while the global implication disappears. The failure resembles neither censorship nor hostility. It resembles a polished executive summary that lets an overloaded institution process more material while noticing fewer discontinuities.

Second-order blindness names the resulting omission. Ordinary review asks whether the claim is credible under present assumptions. It rarely constructs the world in which the claim succeeds and examines the consequences. A research infrastructure may have no users because the relevant users cannot exist before the infrastructure. A technology may fall outside current regulation because its success would make the categories themselves obsolete. Second-order analysis must also include failure, misuse, displacement, concentration, and non-adoption; its purpose is a conditional map, not speculative enthusiasm.

An Longview Advisor should therefore behave like an anti-summary. It should preserve disagreements, separate source claims from inference, label value judgments, expose hidden assumptions, and show which consequences depend on which evidence. Compression should be reversible: the reviewer can move from a conclusion to a scenario, from the scenario to a causal chain, and from the chain to a source, program, dataset, or experiment. The interface should offer selective depth around the few assumptions and consequences that dominate the decision, not generate an encyclopedia no one will read.

There is also a deeper limit that should not be confused with poor explanation. AI may discover a valid regularity, construction, or control policy whose natural representation is too complex for ordinary human intuition. Predictive success is not identical to scientific understanding, and future systems may search spaces that no researcher can traverse directly.

The relevant boundary is among three forms of knowledge. Human-legible science uses concepts and explanations experts can grasp and communicate. Machine-legible science may resist intuitive compression but remains inspectable through executable programs, reproducible transformations, formal invariants, independent tests, and explicit provenance. Machine-opaque claims

provide neither an adequate human account nor a verification path that another system or experiment can challenge. Only the first two can support consequential knowledge claims.

Program-search systems such as FunSearch illustrate the desirable pattern. Candidate discoveries are not accepted as fluent answers; they are expressed as programs and passed through systematic evaluators. The program may not yet constitute a human theory, but it can be executed, scored, inspected, modified, and compared. Intelligibility can sometimes reside in operations and invariants before it can be compressed into a satisfying explanation.

Science should keep trying to translate machine-legible results into human understanding without assuming that nature must fit comfortable human abstractions. Difficulty of explanation must never become authority by opacity. When concepts remain hard, the evidentiary interface must become stronger: reproducibility, independent challenge, boundary tests, counterexamples, and precise statements of what works without a settled account of why.

The Longview Advisor therefore produces two linked outputs: the best human account currently available and the executable evidentiary structure that survives when the account is incomplete. Its purpose is not to make every strange result feel familiar. It is to make the boundary between verified structure, conceptual understanding, and unresolved opacity impossible to hide.

CHAPTER 4

Permission, Institutions, and the Ethics of Delay

4.1 Gatekeepers, Authority, and the Moral Weight of Inaction

Gatekeepers are often described as if their function were inherently suspect. This is mistaken. A research funder must choose among proposals. A regulator must protect people who did not volunteer for an experiment. An editor must allocate scarce review capacity. An investor has fiduciary responsibilities. A hospital cannot deploy every untested intervention. A public institution must explain why one group received resources and another did not. Gatekeeping is one name for decision under scarcity and asymmetric consequence.

The ethical problem begins when a practical mandate is confused with moral sovereignty. A committee member may have authority to allocate a budget, but not superior ownership of the possible futures that budget could enable. A regulator may have authority to restrict deployment, but not an unlimited right to prevent bounded learning. An expert may know more about the current field, but this does not make his preferred ontology the natural boundary of reality.

Most guardians are historically contingent. Their institutions were designed for a particular distribution of risk, pace of change, and form of evidence. They occupy their roles because of law, capital, professional recognition, or organizational inheritance. These are reasons to take their decisions seriously. They are not reasons to treat existing procedures as ethically complete.

The moral asymmetry between action and inaction makes the problem difficult to see. A visible intervention has an identifiable author and outcome. A rejected experiment disappears into the status quo. If the refusal prevents harm, the benefit is hard to observe. If the refusal prevents a benefit, the loss is even harder to attribute. Psychological research on omission and opportunity-cost neglect helps explain why people judge harms caused by action more severely and often fail to represent the alternatives sacrificed by a decision.

Institutions inherit this bias. A regulator is blamed for a harmful product it permits, not for beneficial products that never reach a trial. A funder is criticized for a strange failed project, not for a field that remains stagnant because every project was conventional. A senior manager is punished for an experimental system that creates a public incident, not for ten years of invisible inefficiency. These incentives can make inaction look prudent even when it carries larger expected harm.

The status quo, however, is not a neutral baseline. It is an active distribution of benefits, harms, power, and opportunity. Continuing an inadequate medical process exposes patients to its known limitations. Retaining a slow scientific review system delays error correction. Refusing public experimentation with trustworthy AI can leave deployment to private systems whose objectives are less accountable. Every "no" selects a world, even when the selected world is familiar enough to escape description.

This matters especially during the AI transition. Public institutions may attempt to preserve control by moving slowly. But AI development will not wait for every committee to become comfortable. Actors with compute, data, laboratories, legal teams, and distribution can continue. Actors building open standards, public-interest validation, or unconventional research infrastructure often depend more heavily on permission and

external funding. Indiscriminate caution can therefore select for concentration. It does not prevent experimentation; it changes who is able to conduct it.

The result can be dystopia by default. No institution deliberately chooses a future in which scientific discovery is concentrated, public systems depend on opaque vendors, and promising social technologies die in procurement. Each organization simply applies reasonable rules to unfamiliar proposals. The aggregate outcome is a future shaped by the few actors capable of bypassing those rules.

A morally serious system must account for omission risk alongside commission risk. This does not require numerical certainty about hypothetical benefits. It requires decision procedures that ask what is lost if the proposal is not tested, who bears that loss, and whether delay advantages incumbents. An impact assessment that considers only harms caused by adoption is incomplete. A funding decision that considers only the probability of project failure ignores the value of information and the cost of leaving a bottleneck untouched.

The distinction between belief and experiment is crucial. Gatekeepers are often asked to approve a claim as if the only alternatives were endorsement or rejection. In reality, the ethically appropriate response to high uncertainty is frequently a smaller test. Society need not believe that a radical scientific infrastructure will transform discovery. It may need to permit a controlled prototype that can reveal whether the mechanism works. A regulator need not authorize general deployment. It may authorize a sandbox with monitoring, limited participants, and a stop condition.

This reframing limits the moral power of incredulity. "I am not persuaded" can justify withholding belief. It is not always sufficient to justify preventing inquiry. When a test is cheap, reversible,

consent-based, and capable of generating important information, refusal needs a stronger reason.

Guardians remain necessary, but their role changes. They become stewards of a portfolio of possible futures rather than defenders of one inherited trajectory. Their task is not to eliminate uncertainty before action. It is to decide which uncertainties can be reduced responsibly and which risks must remain bounded until stronger evidence exists.

This is a more demanding ethic than either accelerationism or precaution. Accelerationism can treat resistance as pathology and underestimate irreversible externalities. A crude precautionary approach can treat novelty as a burden imposed on a neutral world and ignore the harms of delay. The appropriate question is comparative and dynamic: which sequence of experiments produces the most responsible learning while preserving the ability to stop?

The answer will vary by domain. A new method of organizing literature can be tested with broad freedom. A self-modifying biological system requires containment and specialized oversight. A financial product that exposes uninformed consumers needs stronger protection than a simulation. Ethical experimentation is proportional. Its urgency does not erase risk; it prevents risk from being represented only on one side of the decision.

The moral claim of this book is therefore not that gatekeepers should surrender. It is that they should lose the presumption that delay is innocent.

4.2 The Duty of Reversible Inquiry

The duty of reversible inquiry is a proposed institutional principle, not an established doctrine. It begins from a simple observation: some experiments have value even when the underlying idea is

unlikely to succeed, because the experiment is cheap, bounded, and informative. Under those conditions, preventing the test can destroy option value at little gain in safety.

A *prima facie* duty to enable inquiry arises when several conditions converge. The possible benefit must be meaningful rather than merely private vanity. The hypothesis must be sufficiently articulated that an observation could change belief. The experiment must have high learning value. Its harms must be bounded, monitored, and reasonably reversible. People exposed to material risk must be protected or provide informed consent. The proposal should not have an obvious lower-risk substitute that yields the same information. These conditions do not guarantee approval, but they reverse the default from "prove the future first" to "show why a responsible test should not occur."

Reversibility is not binary. An experiment can be technically reversible and socially sticky. A pilot algorithm can be removed while data collected during the pilot remains. A research norm can spread beyond the original site. Participants can be harmed by temporary exposure. A responsible design therefore examines technical, institutional, economic, and psychological reversibility. It defines stop conditions in advance and records who has the power to invoke them.

Learning value is equally important. Many pilots are demonstrations designed to succeed rather than experiments capable of changing a decision. They use favorable cases, weak baselines, and metrics chosen after results are known. A reversible inquiry deserves institutional support only when failure remains possible and informative. The experiment should discriminate among live hypotheses, not provide theater for a predetermined conclusion.

The principle resembles permissionless proof, permissioned scaling. Early claims should encounter reality with minimal

ceremonial burden when exposure is limited. Evidence requirements should then rise with scale, irreversibility, and the number of non-consenting parties affected. This creates an evidence ladder. A concept may begin in simulation, move to a sandboxed prototype, then to a controlled field trial, then to limited deployment, and only later to general use. At each stage, the next permission depends on evidence produced at the previous one.

This sequence is ethically superior to demanding production-grade proof before a prototype and strategically superior to deploying at scale on the basis of enthusiasm. It creates a path between paralysis and recklessness.

The principle also supports a right of appeal through evidence. A proposal rejected because it lacks credibility should be able to return with a decisive result without being trapped by the original category judgment. Institutions often treat rejection as a terminal label. A learning-oriented system records the reason for refusal, specifies what evidence would change it, and creates a route for re-entry. This makes gatekeeping falsifiable.

Portfolio design is another consequence. Radical ideas have skewed distributions: many fail, a few create enormous value, and early probabilities are difficult to estimate. Evaluating each proposal as if it must justify itself on average can eliminate the entire class. A portfolio can rationally include projects that would be unreasonable as isolated commitments. Venture capital, ARPA-style programs, scientific philanthropy, and focused research organizations each embody versions of this logic.

Research on scientific incentives supports the importance of institutional design. Azoulay, Graff Zivin, and Manso found that researchers supported through the Howard Hughes Medical Institute's longer-horizon, person-centered model produced more high-impact and more original work than comparable researchers operating under conventional grant incentives, while also

producing more failures. Manso's broader theory of innovation incentives similarly emphasizes tolerance for early failure and reward for long-run success. The relevant lesson is not that one funding model should replace all others. It is that evaluation rules change what kinds of search people are willing to undertake.

An institution committed to reversible inquiry would therefore measure more than successful outputs. It would measure whether experiments reduced important uncertainty, whether the portfolio preserved methodological diversity, whether negative results were reusable, and whether rejected proposals received clear evidentiary routes back into consideration. Failure without learning would remain failure. Failure that closes an expensive branch of possibility would be recognized as an asset.

AI makes this principle implementable at greater scale. Systems can help decompose proposals, identify assumptions, design discriminating tests, simulate failure modes, monitor pilots, and compare results across a portfolio. They can also make the process worse by generating bureaucratic documentation faster than anyone can inspect it. The objective should not be automated permission. It should be automated preparation for responsible experimentation.

The Longview Advisor proposed in the next chapters is designed around this objective. Its most valuable output is not an impact score. It is a test path: the smallest sequence of observations capable of changing a decision while preserving safety. A high-uncertainty proposal with a strong test path can be more actionable than a moderately plausible proposal whose assumptions cannot be examined.

The duty of reversible inquiry also imposes obligations on innovators. They must not treat the language of historic opportunity as a substitute for experimental design. They must expose assumptions, define failure, identify externalities, and

accept staged evidence requirements. The moral criticism of gatekeeping is strongest when the innovator has done the work required to make a bounded test possible.

This reciprocity matters. Institutions owe unfamiliar ideas a fair route to evidence. Innovators owe institutions an experiment that can teach rather than merely persuade.

The resulting ethic is demanding but not utopian. It does not require consensus about the future. It requires agreement that under accelerating change, refusing to learn can be as consequential as learning badly.

CHAPTER 5

Longview Advisor: A Technology for Future-Potential Review

5.1 Users, Decisions, and the Longview Dossier

Longview Advisor should be understood as a technology for decision-makers, not as a metaphor for good judgment. Its primary users are grant reviewers, scientific programme managers, philanthropic funders, investors, R&D leaders, and other temporary gatekeepers who must decide whether an unfamiliar idea deserves attention, money, permission, or a bounded experiment. Longview widens the field of view without transferring the decision itself to the machine.

The product is intentionally adviser-like, but it should be the opposite of an oracle epistemically. It exposes uncertainty, alternative worlds, evidence gaps, and value choices; it records why an implication was generated; and it must remain challengeable. Its value lies in making hidden possibility inspectable, not in claiming privileged access to the future.

Longview is valuable only if its users and decisions are defined narrowly enough to test. The primary user is not "humanity" and the product is not a general futurist companion. The first user is a person who must allocate attention, permission, or money under uncertainty and who cannot personally reconstruct every future opened by every unfamiliar proposal. That includes grant reviewers, scientific programme managers, philanthropic funders, venture investors, corporate R&D leaders, research-infrastructure managers, and innovation teams inside public agencies.

Grant reviewers are the clearest initial use case because the decision has an explicit source document, an explicit rubric, a declared funding context, and an observable downstream outcome. Review is also under measurable pressure. The rapid rise in submissions increases workload, while studies of peer review continue to find substantial reviewer disagreement. Pier and colleagues found very low agreement when the same NIH applications were evaluated by different reviewer groups. Bol and colleagues showed how small early differences in funding can compound into much larger later funding advantages. These findings do not make peer review irrational; they show why a decision-support layer should expose uncertainty rather than convert it into false precision.

For a grant reviewer, Longview would run beside the authorised review process. It could reconstruct the proposal without author identity during an idea-first stage, retrieve a portfolio of supporting and contradicting evidence, identify whether the claimed novelty is mechanistic or mainly rhetorical, and produce several conditional success worlds. It would then ask which assumptions dominate the upside and which small experiment would most efficiently discriminate among them. The reviewer would see the result as a second opinion and remain responsible for the decision. In regimes where funder rules prohibit AI-assisted merit assessment, the technology must not be used for that purpose. The European Research Council, for example, clarified in March 2026 that reviewers may not use AI to summarise proposals, assess scientific merit, or generate draft evaluations. That constraint is not a footnote; secure and authorised deployment is part of the research problem.

Programme managers have a different need. They care about portfolios, not just projects. A proposal can be worth funding because it opens infrastructure that several other projects need,

because it tests a bottleneck common to a field, or because it provides diversity against a dominant technical path. Longview can represent these relationships explicitly. Instead of asking only which proposal has the highest expected score, the programme manager can inspect option value, redundancy, complementarities, shared failure modes, and the value of preserving multiple approaches until evidence deserves convergence.

Investors and business angels face an analogous but not identical decision. A venture investor needs to understand technical possibility, market formation, timing, value capture, regulation, and founder execution. Conventional due diligence is excellent at many of these tasks once a category has formed. It is weaker when the category itself is the thesis. Longview can create a future-potential review that says not simply whether the startup looks investable today, but what must become true for the category to matter, which adjacent technologies it depends on, which actors gain or lose bargaining power, what new markets become possible, and which milestone collapses the greatest uncertainty.

The investor lens must not silently replace public-interest evaluation. A technology can be extremely valuable to an investor while increasing systemic fragility, surveillance, concentration, or externalised harm. Longview should therefore support at least two concurrent lenses: the declared interests of the decision-maker and a broader stakeholder or public-interest lens. The system does not need to pretend that moral weights are objective. It needs to make clear when a recommendation changes because the value frame changed rather than because the evidence changed.

Corporate R&D leaders can use Longview for internal portfolio review. Large organisations often reject strange projects because they do not fit current business units, metrics, or incentive structures. A prospective review can distinguish "not strategically relevant under today's organisation" from "technically irrelevant"

and can make explicit whether the main barrier is science, integration, regulation, or internal cannibalisation. This matters because many transformative technologies initially threaten the structure that must decide whether to fund them.

Philanthropic funders and science patrons may be among the most natural early adopters. They can finance public goods, validation infrastructure, neglected fields, and high-option experiments that lack an immediate market. Longview can help them compare projects whose benefits are diffuse or long-term and whose value cannot be captured by a company. The system's stakeholder and axiological layers are especially important in this context because the funder's mission is often explicitly normative.

Finally, innovators themselves can use Longview before approaching a gatekeeper. This is not the primary governance use case, but it is a valuable secondary one. A founder or scientist can ask the system to identify where the proposal depends on reputation, where the future story outruns the evidence, which claims are likely to be misunderstood, and which demonstration would make the idea legible without diluting it. Marketing then becomes part of epistemic design: not persuasion at any cost, but routing the right decision-maker toward the right evidence.

Table 2 · Users and the question Longview adds

The output must be useful to these users without becoming another forty-page report. The core artifact is a Future-Potential Review, presented through a compact Longview Dossier. Its first screen or first page should tell the reviewer what they need to know before deciding whether deeper reading is warranted. It states the strongest faithful version of the proposal; the central mechanism; the most important plausible future unlocked by success; the most important reason to doubt that future; the key assumptions; the value and stakeholder sensitivities; and the next experiment that would most efficiently change the decision.

Underneath this orientation layer, the dossier becomes navigable. Each implication is marked by epistemic role. A statement may be a source claim, a directly retrieved fact, an inference produced by Longview, a scenario assumption, an analogy, a value judgment, or a human reviewer annotation. The difference is visible. The system should make it difficult for an attractive inference to masquerade as a fact simply because it appears in the same paragraph.

The dossier should also preserve alternative interpretations. Frontier proposals often use new vocabulary, overloaded concepts, or terms whose meaning is still unstable. An ordinary summary tends to choose one interpretation and continue. Longview should instead create branches when the interpretation materially changes downstream implications. A reviewer can then see that the proposal is strong under one reading and weak under another, which often leads directly to a clarification request or a decisive experiment.

The future-potential layer is not a single optimistic scenario. At minimum, the system should explore narrow success, broad success, technical success with institutional failure, partial reuse of infrastructure, adversarial or harmful adoption, displacement by a better alternative, and an ordinary failure world. These branches are not forecasts with percentages attached for theatrical effect. Their purpose is to reveal dependencies and option value. A proposal that creates a reusable verification layer may matter even if its original product thesis fails. A proposal that only succeeds under a fragile regulatory assumption may be much less valuable than its technical review suggests.

The dossier should explicitly contain a counter-case written by an independent critic. The critic's job is not to be negative; it is to identify the most economical explanation under which the proposal is unimportant. It asks whether the novelty is merely linguistic, whether existing methods already recover most of the value,

whether the proposed mechanism is unnecessary, whether success would be absorbed by incumbent infrastructure, and whether the supposed future can be reached through a simpler path. A good Longview review is one in which the optimistic and skeptical branches both become sharper.

The final element is an experiment portfolio rather than a verdict. The system ranks candidate tests by information value, cost, reversibility, time, and ability to distinguish competing explanations. For a software technology, the next step might be a benchmark against a strong baseline or a blinded user study. For a scientific claim, it may be replication, a targeted measurement, or a simulation with predeclared failure criteria. For an institutional innovation, it may be a sandbox or randomised trial. The preferred outcome is not "fund" or "reject" but a route such as "small exploratory grant," "qualified-lottery pool," "independent validation," "wildcard consideration," "continue with explicit safety constraint," or "insufficiently specified to test."

The fact that Longview uses the interests of the reviewer does not make it a personalised persuasion machine. A lens profile is a declared contract. It can specify a funder's mission, an investor's return horizon, a public agency's strategic priorities, or a research programme's scientific objectives. The system then shows how those priorities influence relevance. It should also run sensitivity analysis: if modest changes in the value weights reverse the conclusion, the dossier reports a contested value judgment rather than a stable recommendation.

This is the product proposition in its simplest form. Longview Advisor does not promise to know the future. It gives a reviewer a better way to inspect the futures that become plausible if an unfamiliar idea works, and to convert that inspection into a bounded next action.

Decision context	Longview contribution
Grant review	A conditional account of what a proposal could unlock, why that matters, what is doubtful, and which bounded test most improves the decision.
Investment	A future-potential map connecting mechanism, category formation, dependencies, market and institutional consequences, and the milestone that collapses the largest uncertainty.
Research programme management	Portfolio-level option value, complementarities, shared bottlenecks, diversity, and event-triggered continuation decisions.
Philanthropy and public-interest funding	Long-horizon and public-good implications that may be poorly represented by immediate market signals, with explicit stakeholder and value lenses.
R&D and strategic technology	Separation of scientific possibility from organisational fit, allowing a company or institution to see opportunities that threaten present structures.

5.2 Second-Order Review and Why Longview Cannot Be Only a Chatbot

Imagine a reviewer receiving a proposal that fits no established category. The reviewer has forty minutes, limited domain knowledge, and a template designed for conventional projects. A chatbot can summarize the text, list risks, compare related work, and recommend questions. This is useful. It does not perform the missing operation.

The missing operation is prospective reconstruction: building the worlds in which some part of the proposal becomes true. What

must exist around it? Which constraints disappear and which bottlenecks emerge? Who gains a new option, who loses power, and what becomes cheaper, faster, safer, or more concentrated? What evidence would show whether the proposal opens a field or merely renames an old mechanism?

The Longview Advisor is a second-order review system for this task. It does not predict one future or return a universal impact score. It constructs conditional futures and makes their dependencies, evidence, values, uncertainty, and experiments inspectable. Its central question is: if this works, what follows, for whom, under which assumptions, and what should be tested next?

The system begins with three inputs: the source, the decision context, and a lens profile. The source may be a proposal, paper, patent, book, technical design, or recorded explanation. The decision context may be funding, peer review, regulation, acquisition, partnership, or research prioritization. The lens profile describes the user's objectives and constraints, but it does not silently become the moral universe of the report. Relevance to the user, effects on stakeholders, and broader public implications remain separate.

Processing begins with faithful reconstruction. The engine identifies the mechanism, intended outcome, evidence, scope conditions, unstated prerequisites, and claims that are rhetorical rather than testable. Ambiguity produces alternative interpretations instead of a convenient single reading. Novelty analysis then distinguishes new language from new combinations, mechanisms, objectives, and institutional forms, using prior art, adjacent disciplines, failed attempts, and recent enabling developments.

Counterfactual expansion constructs a small set of deliberately different worlds: obvious failure, narrow success, technical success with institutional failure, scalable success, infrastructural reuse,

adversarial adoption, and non-adoption while an inferior alternative becomes dominant. These are not forecasts with theatrical precision. They are stress surfaces that make hidden dependencies and neglected consequences visible.

Causal, stakeholder, and value analysis decomposes broad claims into pathways and prerequisites. It identifies beneficiaries, burdens, displaced roles, power shifts, distributional effects, misuse, and conflicts among welfare, autonomy, fairness, resilience, openness, privacy, pluralism, and environmental impact. Option-value analysis asks whether a standard, dataset, instrument, verification layer, or open platform matters by preserving future choices or enabling an ecosystem even when direct value capture is weak.

The final analytical stage designs evidence. The engine identifies assumptions with the greatest influence on the decision and proposes the smallest tests capable of changing belief: a strong-baseline benchmark, replication, user study, sandbox, formal check, red-team exercise, market test, simulation, or instrumented pilot. It prefers tests that remain informative when the project fails.

The output is a Decision Dossier. Its first page states the strongest faithful form of the idea, the most consequential plausible implication, the principal reason for doubt, the highest-leverage assumptions, relevance to the user, serious externalities, and the next decisive test. Deeper layers expose the reasoning graph, evidence, scenarios, evaluator disagreement, and the recommended institutional route. The engine succeeds not when it sounds prophetic, but when it reveals a consequential possibility or risk that ordinary review missed and connects that insight to an action.

Table 1 · Ordinary review and second-order review

The Longview Advisor is deliberately not called a futurist oracle. Foresight is not prediction. Serious foresight explores plausible

alternatives, contextual forces, and decision robustness. The system should be rewarded for revealing decisive uncertainty, not for producing confident stories. It succeeds when the reviewer sees a possibility or risk that ordinary review would have missed and can connect that insight to an action.

The strongest use case may be the proposal that initially appears strange but contains a tractable test. The engine can convert cultural distance into an experimental path. This is how it helps break the permission paradox: it does not ask the gatekeeper to believe the future. It shows the gatekeeper how to investigate it.

A chatbot is a conversational interface over a model. The Longview Advisor is an evaluation system with persistent state, independent workstreams, tools, explicit representations, and a reproducible process. The distinction determines whether the output is a persuasive answer or an inspectable decision object.

First, the engine organizes information by epistemic role. A proposal, literature, code, reviewer comments, user priorities, and scenario branches cannot be managed reliably as one undifferentiated conversation. The system records source claims, reconstructions, assumptions, evidence, values, scenarios, experiments, and the agent or person responsible for each inference. When the proposal or evidence changes, affected conclusions can be recomputed rather than the entire account regenerated from memory.

Second, it treats independence as a design and measurement problem. One model generating, criticizing, and judging its own work can simulate diversity without acquiring external evidence. The engine separates roles, model families, evidence sets, and evaluation methods; permutes candidate order; hides reputation during idea-first analysis; and measures correlated disagreement. Factual claims are routed to tools or primary sources, while

consequential weakly grounded conclusions trigger human escalation.

Third, it enforces evidence discipline and scenario persistence. Citations are checked for entailment and common ancestry; direct evidence, analogy, expert opinion, and system inference remain distinct. Scenarios persist as objects with assumptions, causal links, actors, indicators, and branch conditions. A recommendation can therefore be traced to the evidence and values that produced it and compared under alternative lenses.

Fourth, the engine closes the loop with experiments. It does not end with advice. It specifies what should be measured, the baseline, intervention, stopping rule, confounders, evidentiary threshold, and result that would change the recommendation. In computational domains it can emit executable code, simulations, or data transformations. In physical domains it produces machine-readable protocols for expert and laboratory review.

Fifth, the process is repeatable and evaluable. Model versions, retrieval results, policies, transformations, human interventions, and report changes are recorded. Historical benchmarks use temporal cutoffs and include eventual successes, failures, harms, and inconsequential novelties. Prospective trials compare ordinary and engine-assisted review on fidelity, calibration, traceability, non-obvious implication coverage, usefulness of proposed tests, appropriate changes after new evidence, and whether the system improves judgment rather than merely confidence.

Sixth, the interface resists semantic closure. A compact decision page is accompanied by visible disagreements, unsupported branches, missing stakeholders, and unresolved assumptions. The evaluator can move from conclusion to scenario, causal chain, claim, source, or experiment. The aim is not maximum text but selective depth at the points where attention can change the decision.

Finally, the engine belongs inside the research state rather than beside it. In *ex ante* mode it asks whether an idea deserves a test and which test is decisive. In *active* mode it connects code, protocols, data, observations, negative results, and revised hypotheses to a living evidence state, triggering human review when an assumption fails, a safety boundary changes, a branch requires resources, or accumulated failure demands a new representation. In *ex post* mode it separates surviving claims from abandoned ones and exports reusable artifacts and negative knowledge.

Continuous review must not become continuous surveillance or automated sovereignty. Most updates can be checked mechanically for provenance, completeness, and consistency, while scarce human attention is reserved for value conflicts, tacit knowledge, ambiguity, appeals, and decisions beyond the system's mandate. Dedicated secure infrastructure, controlled models, auditable policies, and named human accountability are prerequisites; uploading confidential proposals to an external chatbot is not the system described here.

The difference can be stated simply. A chatbot generates an answer about an idea. The Longview Advisor maintains an argument about the idea that can be inspected, contested, tested, and updated. Because the system will itself hallucinate, smooth, and miss tacit structure, its most important property is not apparent intelligence but the ability to make its own uncertainty and mistakes leave traces.

Ordinary review	Second-order review
Asks whether the proposal is correct, feasible, novel, and acceptable under current categories.	Asks what new capabilities, institutions, risks, and options appear if important parts of the proposal work.

Ordinary review	Second-order review
Produces a verdict, score, ranking, or recommendation.	Produces a conditional map, an uncertainty ledger, and a sequence of decisive tests.
Uses present evidence and established comparators as primary anchors.	Uses present evidence but also examines missing comparators, enabling dependencies, and category-changing effects.
Treats the user's decision objective as the main frame.	Separates user relevance, stakeholder consequences, and broader public value.
Compresses disagreement into a final judgment.	Preserves disagreement when it corresponds to genuinely different assumptions, values, or causal models.
Explains why the proposal should or should not proceed.	Explains what would change the decision and how to learn it responsibly.

CHAPTER 6

How Longview Could Be Built

6.1 An LLM-Heavy Architecture

The fastest implementation of the Longview Advisor would be built around language models because they are unusually capable at reading heterogeneous documents, translating across domains, generating hypotheses, and producing explanations for different audiences. The architecture should exploit these strengths while refusing to treat fluent model output as a final epistemic object.

The system begins with an intake compiler. It accepts the source material, supporting files, decision context, and lens profile. The compiler segments the source and asks several models to reconstruct its central mechanism independently. Differences among reconstructions are not immediately averaged. They become an ambiguity set presented to the user or routed to a resolving stage. This prevents the first interpretation from silently defining the rest of the analysis.

The compiler classifies statements by function. It separates empirical claims, causal claims, engineering requirements, forecasts, analogies, definitions, value claims, and rhetorical framing. It identifies terms whose meaning appears local to the proposal and creates a glossary linked to source passages. It records what the author asserts, what is implied, and what remains unspecified.

The second component is an evidence service. It combines keyword search, semantic retrieval, citation graphs, patent search, project databases, standards, technical documentation, and domain-specific repositories. Retrieval is organized around claims and mechanisms rather than the source document as a whole. A novelty agent searches for equivalent mechanisms under different

terminology. A contradiction agent searches for evidence that would defeat the central assumptions. A historical agent looks for earlier attempts, including failures that may have become newly relevant because enabling conditions changed.

The evidence service should privilege primary sources for factual support and record lineage among sources. Ten articles discussing the same experiment are not ten independent confirmations. The system can cluster evidence by origin, note retractions or disputes where known, and distinguish absence of evidence from evidence of absence.

The third component is a panel of genuinely differentiated evaluators. A scientific-validity agent examines mechanisms and evidence. An implementation agent asks what would be required to build and operate the system. A systems agent examines dependencies and bottlenecks. An adoption agent studies incentives, workflow change, and diffusion. An institutional agent asks which organizational forms could carry the idea. A security and misuse agent explores adversarial paths. An axiological agent identifies value conflicts and affected groups. A historical analogist examines patterns without treating analogy as proof.

The value of the panel depends on heterogeneity. Different model families can be used for consequential roles. Some agents can receive blinded source metadata to reduce prestige effects. Some can be given only the claim graph and evidence; others can inspect the author's narrative. Evidence sets can be varied deliberately to measure sensitivity. An orchestration layer records where conclusions converge because of shared facts and where they converge merely because the agents share a linguistic prior.

The fourth component is a scenario workbench. It combines horizon scanning, futures wheels, cross-impact analysis, morphological exploration, backcasting, and adversarial scenario design. The workbench generates a small set of structurally

different futures rather than many stylistic variants. Each scenario has explicit assumptions, triggers, causal pathways, beneficiaries, harms, indicators, and failure conditions.

The scenario process should begin with invariants and branching variables. Some consequences follow in almost every plausible world; others depend on adoption, regulation, complementary technologies, or strategic behavior. The engine highlights robust implications separately from branch-specific speculation. It also includes an ignore scenario, because the consequences of non-adoption are central to the ethics of delay.

The fifth component is an interest and value lens. The requesting organization's objectives are represented explicitly, but the engine also runs stakeholder and public-interest lenses that cannot be edited into disappearance. A company may assign high importance to strategic fit and defensibility; the report can still show that the same pathway centralizes power or transfers risk. The system should never transform a preference into a fact.

The sixth component is an experiment planner. It takes the most influential assumptions and searches for observations that maximize decision-relevant learning. Language models can propose tests, but external tools should evaluate feasibility. Code-generating agents can implement computational experiments in sandboxes. Statistical modules can inspect power, baselines, and confounders. Policy pilots can be checked against legal constraints. Human experts review any experiment involving physical, medical, security, or environmental risk.

The seventh component is a calibration and adjudication layer. LLM-as-judge methods can be useful for comparing hypotheses, but known biases make a single judge unacceptable. The system should use blinded pairwise comparisons, order permutations, multiple judges, reference-based checks where possible, and uncertainty estimates derived from disagreement and evidence quality rather

than model confidence alone. A meta-reviewer synthesizes but cannot erase unresolved dissent.

The eighth component is the dossier renderer. It produces several synchronized views. The decision page presents the central implication, strongest reason for doubt, relevance to the user, major externality, and next test. The scenario view displays conditional futures. The evidence view traces claims to sources. The value view shows whose interests drive each recommendation. The experiment view defines the learning path. The dissent view preserves substantive disagreement.

The entire workflow is represented as a directed process rather than a chat transcript. Tasks can be rerun when sources change. Intermediate artifacts are versioned. The model router can choose fast systems for extraction and stronger systems for high-leverage reasoning. Code execution and retrieval occur in controlled environments. Every output carries provenance sufficient for audit.

A practical minimum version does not require every component. It can begin with document ingestion, claim decomposition, evidence retrieval, three independent scenario agents, a stakeholder lens, an adversarial reviewer, and an experiment planner. The crucial step is architectural: outputs must be stored as structured, addressable objects, not allowed to dissolve into one final answer.

The LLM-heavy architecture has compelling advantages. It can be built quickly, works across domains, handles incomplete language, and discovers connections that a fixed ontology may miss. It also has serious weaknesses. Its agents may share hidden biases, scenario generation can become verbose fiction, evidence synthesis can overgeneralize, and adjudication can reward rhetorical quality. The architecture therefore needs external anchors and evaluation from the beginning.

The most important benchmark is temporal. Historical proposals can be reconstructed at the moment before recognition. The engine

receives only sources available at that time and is asked to identify implications and experiments. Cases should include eventual successes, failures, harms, and ideas that were novel but irrelevant. This prevents the benchmark from becoming a museum of misunderstood geniuses.

Prospective evaluation is equally necessary. Live proposals can be randomly assigned to ordinary review or engine-assisted review. Independent experts can measure whether the engine changes attention, reveals important assumptions, improves experiment selection, or merely increases confidence and report length. The system should be considered harmful if it makes reviewers more certain without improving calibration.

An LLM-heavy Longview Advisor is feasible now. Whether it becomes valuable depends less on model eloquence than on the discipline of the surrounding workflow.

6.2 A Neuro-Symbolic Architecture and an Executable Review Stack

A neuro-symbolic architecture starts from a different concern. Language models are powerful generators and interpreters, but the object being built is not primarily a conversation. It is a structured theory of a proposal: what claims it contains, how they depend on one another, what evidence supports them, what consequences follow under different assumptions, and which tests can change the state of the theory.

The central artifact is a typed implication graph. Its nodes represent claims, mechanisms, prerequisites, resources, actors, institutions, values, evidence items, risks, experiments, and outcomes. Its edges represent relations such as supports, contradicts, requires, enables, causes, substitutes, benefits, harms, constrains, measures, and

falsifies. Every node and edge carries provenance, scope, uncertainty, and the agent or person responsible for introducing it.

A language model serves as a semantic intake layer. It maps passages into candidate graph elements and proposes interpretations. The symbolic kernel does not accept these silently. It checks type constraints, detects duplicate or incompatible concepts, and asks for clarification where the mapping changes downstream conclusions. Human corrections become part of the trace rather than disappearing into a revised paragraph.

The graph is actually several linked graphs. The claim graph represents the argument. The evidence graph records support, contradiction, source lineage, and replication. The causal graph represents mechanisms and interventions. The stakeholder graph represents actors, incentives, and power. The value graph represents objectives and trade-offs. The experiment graph represents possible tests, observables, stopping rules, and the beliefs each result would affect.

This separation prevents a common failure of prose. A statement can be causally plausible, weakly evidenced, valuable to one actor, harmful to another, and difficult to test. In a summary these properties merge into tone. In a structured representation they remain distinct.

The symbolic kernel applies several families of operations. Argumentation operators propagate support and attack while preserving cycles and unresolved conflicts. Constraint operators test whether a scenario requires mutually incompatible assumptions or unavailable resources. Causal operators examine interventions and counterfactuals. Backcasting starts from a high-impact outcome and identifies necessary intermediate states. Morphological operators combine alternative values of key variables to generate structurally different scenarios. Multi-criteria

operators compare implications under explicit value weights without claiming that the weights are objective facts.

Formal methods can be used where the proposal contains formalizable components. Software properties can be checked against specifications. Data transformations can be replayed. Protocol steps can be validated for completeness. Policy constraints can be encoded as rules. The architecture should not pretend that social adoption, scientific significance, or moral value can be proved mathematically. Symbolic precision is applied to the parts that deserve it, while uncertainty remains explicit elsewhere.

The system uses abduction as well as deduction. It asks what must be true for the proposal's claimed outcome to occur and generates candidate explanations for gaps. These abductive hypotheses are labeled and routed to evidence search or experiment design. They are not promoted to facts because they make the graph coherent.

The interest lens becomes a view over the value and stakeholder graphs. A user can specify objectives, constraints, and time horizon. The system then computes which pathways are relevant and why. A public-interest lens can run in parallel. If the recommendation changes dramatically under modest value adjustments, the dossier reports ethical sensitivity rather than a stable verdict.

The experiment compiler is particularly important. It identifies high-centrality assumptions whose truth controls many downstream implications. It then searches for tests that distinguish competing graph states. A computational test can be emitted as an executable task. A physical experiment can be represented as a protocol with required instruments, controls, observables, and review checkpoints. Results update only the connected claims, allowing the report to evolve rather than be rewritten from scratch.

The neuro-symbolic architecture has advantages that language-only systems struggle to provide. It is traceable, repeatable, and capable of preserving contradiction without smoothing. It can expose

hidden dependencies, compare scenarios systematically, and keep user values separate from empirical claims. Its outputs are easier to audit, and its state can connect directly to an executable-science environment.

Its weaknesses are equally real. Ontologies can freeze the categories of the present. Natural-language proposals contain ambiguity, metaphor, tacit knowledge, and genuinely new relations that a fixed schema may distort. Knowledge engineering is expensive. A graph can create an illusion of rigor if uncertain edges are represented with precise-looking labels. Symbolic systems can be internally consistent and externally irrelevant.

For these reasons, the recommended architecture is hybrid. Language models provide breadth, interpretation, analogy, and candidate generation. The symbolic layer provides memory, invariants, provenance, explicit values, scenario structure, and controlled updates. External tools provide empirical contact. Humans provide domain judgment, accountability, and the ability to recognize when the representation itself is wrong.

The two architectures should not be treated as rival doctrines. The LLM-heavy design reaches useful breadth earlier; the neuro-symbolic design makes repeated, consequential evaluation more governable. The practical question is which operations require generative flexibility and which require persistent structure. A credible system begins with language-model orchestration, then progressively moves claims, values, decisions, and experimental state into typed objects as the domain and stakes justify the investment.

A hybrid system can be organized as a loop. Language models parse the proposal and generate candidate implications. The symbolic kernel records and tests them. Evidence services support or attack them. Scenario operators expose conditional consequences. Experiment planners convert uncertainty into tests. Results update

the graph. Language models then explain the revised state to different audiences without changing the underlying record.

This loop is more than an elaborate answer to a prompt. It is a small institutional memory for possibility.

The architecture also aligns naturally with executable science. A claim node can link to a source span, dataset, code artifact, protocol, review comment, and replication. An axiological claim can link to the stakeholder and value assumptions on which it depends. A reviewer can see not only that the system recommends a pilot, but which chain of evidence and values produced that recommendation. An agent can invoke a computational experiment and return the result to the same graph.

No implementation should attempt universal coverage at the beginning. A credible path starts with a narrow domain in which claims and experiments can be represented well, then expands through modular ontologies and adapters. Scientific review, technology due diligence, and research-program design are plausible initial domains because they already contain explicit documents, evidence standards, and decision milestones.

The system should also support plural evaluators. Different organizations can run their own value lenses and scenario policies over a shared evidence graph. Open standards can allow dossiers to be exchanged and contested. A centralized implication oracle would reproduce the permission paradox at a new level. The objective is an ecosystem in which evaluations are interoperable, inspectable, and capable of disagreement.

The best technical principle for the Longview Advisor is therefore the same as the social principle of this book: preserve variation until evidence deserves convergence.

CHAPTER 7

Funding the Epistemic Clock

7.1 Smaller Grants, Shorter Cycles, Longer Patience

The Discovery Flood changes not only how research is performed but the natural size and duration of a grant. When AI can compress literature analysis, software construction, simulation, data exploration, and experimental planning, institutions should not continue assuming that every serious idea needs a three-year narrative before it may encounter reality.

The conventional grant is partly a product of transaction costs. Calls are infrequent, proposals are expensive to prepare, reviews are slow, contracts are difficult to amend, and administrative reporting consumes substantial effort. Because every funding decision is costly, institutions bundle many uncertain activities into one large commitment. Applicants must describe in advance what they will discover, divide that imagined future into work packages, and promise deliverables precise enough to administer but vague enough to survive contact with research.

AI reduces some of these transaction costs, but it does not make every field fast. Software experiments may produce decisive evidence in days. A materials synthesis may require months. A clinical outcome, ecological process, telescope campaign, or new scientific instrument may have an irreducible clock measured in years. Funding should therefore follow the epistemic clock: the minimum interval after which reality can provide information capable of changing the decision.

The future grant may be less a contract predicting several years of research and more a renewable option to purchase the next

decisive piece of evidence. The first commitment finances mechanism clarification, a benchmark, a minimal instrument, a replication, or a bounded experiment. The next commitment is released when the result justifies a larger question. A project can stop cheaply, branch into competing explanations, or receive substantially more capital when uncertainty collapses in its favor.

This model produces smaller and shorter grants in many AI-accelerated domains, but it should not produce short-termism. The appropriate formula is short cycles and long patience. A difficult program may continue for a decade through a sequence of evidence-bearing commitments. It is not forced to manufacture annual triumphs, yet neither does its historical ambition exempt it from showing what each cycle taught.

The European Innovation Council's 2026 Advanced Innovation Challenges pilot points in this direction. Its first stage offers a EUR 300,000 lump sum for up to nine months to prepare, validate, and benchmark breakthrough solutions; a later stage can provide up to EUR 2.5 million for the most promising projects. This is not a complete institutional model, but it demonstrates that European funding can separate rapid evidence creation from larger development commitments.

Real science must be able to fail. It need not fail in every project, and failure is not automatically virtuous. The essential condition is that the hypothesis can lose. A program in which every milestone is retrospectively renamed a success is not protected exploration; it is a reporting system that has made itself unfalsifiable.

The relevant distinction is between productive failure and drift. Productive failure changes the search. It falsifies a mechanism, reveals a constraint, improves an instrument, produces a reusable negative result, or shows that the problem was represented incorrectly. Drift consumes time while preserving every hypothesis and repeatedly promising that the decisive evidence is just beyond

the next extension. Institutions should be tolerant of the first and much less tolerant of the second.

A Failure Ledger makes the distinction operational. For each material attempt, it records the hypothesis, intervention, observation, expected discriminator, result, uncertainty reduced, artifacts produced, and the branch now closed or opened. The ledger is not a register of shame and should not be used as a simplistic performance score. It is scientific memory. Under AI, its importance increases because thousands of agents may otherwise rediscover the same dead ends in fluent new language.

Periodic review is poorly matched to this structure. A review scheduled at month twelve or month eighteen may occur when nothing decisive has happened, while an important contradiction discovered a week later waits until the next formal cycle. The result is ceremonial accountability: dense presentations, reconstructed narratives, and documents optimized for the review rather than for the research.

The alternative is a living evidence state. The proposal, claims, protocols, code, data transformations, experimental runs, failures, decisions, and revisions are connected continuously. Longview Advisor can verify whether an artifact exists, whether a claim changed, whether evidence supports the declared milestone, and whether the current direction still corresponds to the funded rationale. Human review is triggered by epistemic events, not by the arrival of a date.

Such events include a central assumption being falsified, a branch requiring substantially more resources, a safety boundary being approached, a result that makes the original plan obsolete, or a sequence of failures that suggests a different conceptual framework. Routine evidence accumulation should not summon a panel. Consequential changes should.

This architecture also makes lump-sum funding easier to validate. The meaningful question is not whether every expense was placed in the correct historical category but whether the agreed evidence-producing activity occurred and what it taught. Code, data, protocols, experiment records, and independent checks provide a stronger basis for a tranche decision than a polished narrative about percentage completion. Financial audit remains necessary where public money is involved, but it no longer needs to impersonate scientific evaluation.

AI should make research bureaucracy shrink as fast as scientific possibility expands. The opposite outcome is entirely possible. Institutions may use language models to generate longer work packages, more elaborate risk tables, automated progress prose, and synthetic reviewer comments while preserving every old ceremony. That would be automation of bureaucracy rather than automation of science.

Caution about AI-assisted evaluation is legitimate. In March 2026, the European Research Council clarified that reviewers may not use AI tools to summarize proposals, assess scientific merit, or draft evaluations, and may not upload confidential applications to external systems. Given present confidentiality, accountability, and model-governance problems, this restriction is reasonable. The system proposed here is not an invitation to paste applications into a public chatbot. It requires dedicated infrastructure, controlled models, secure processing, provenance, auditable policies, and a human decision-maker who remains responsible.

The institutional gain is not that machines decide continuously. It is that evidence becomes continuously available while human attention becomes more selective. Machines can check provenance, execute tests, detect inconsistencies, maintain the argument graph, and identify triggers. Humans adjudicate competing values, tacit

domain knowledge, irreducible ambiguity, and changes whose consequences exceed the project's mandate.

A grant system organized around the epistemic clock can be faster without becoming impatient. It can stop weak branches earlier, renew difficult work for longer, and spend less on pretending that discovery follows a calendar written before the experiment began.

7.2 Reputation Firewalls, Qualified Lotteries, and Portfolio Design

When evaluation uncertainty is high, reputation becomes a substitute for understanding. The substitute is often useful, but it is also cumulative: prior funding, institutional prestige, famous collaborators, and visible endorsements influence who receives the next opportunity, which then creates the evidence used to justify further reputation.

The first reform is sequencing. The question whether an idea deserves an experiment should be separated from the question whether a particular team can execute it. In an idea-first stage, names, institutions, prior investors, citation counts, and other status signals are hidden unless they are part of the mechanism itself. In a later execution stage, the evaluator examines competence, access to infrastructure, governance, and delivery risk. This is not full anonymity and cannot remove every clue. It is a reputation firewall that prevents status from answering the wrong question too early.

Longview Advisor can enforce the separation technically. The initial run receives a sanitized source package and produces a mechanism reconstruction, novelty map, implication analysis, uncertainty ledger, and proposed tests. Identity and track-record information live in a distinct capsule. Only after the idea crosses an epistemic

threshold does the system estimate execution risk. The final dossier shows which conclusion changed when reputation became visible.

The second reform concerns proposals that pass a threshold but cannot be ranked reliably. Here a qualified lottery is more honest than theatrical precision. Reviewers first exclude work that is incoherent, unethical, irrelevant to the program, or incapable of producing informative evidence. All proposals that meet the published threshold enter a randomized allocation for the funding reserved to that lane.

This is not an abstract provocation. New Zealand's Health Research Council uses anonymous assessment and random ordering for eligible Explorer Grant proposals judged transformative and viable. The Swiss National Science Foundation allows lots when its funding line cuts through proposals of indistinguishable scientific quality. The British Academy has extended partial randomization of its Small Research Grants to 2028 after reporting a broader pool of applicants, awardees, and represented institutions. Austria's FWF concluded in 2026 that its 1000 Ideas program is strongest as a high-risk, high-gain format with small budgets, short terms, anonymous assessment, and the possibility of partial randomization.

These examples do not prove that lotteries generate more breakthroughs than every alternative. Long-run comparative evidence remains limited, and public trust can be damaged if randomization is presented as indifference to quality. Their importance is narrower and still substantial: qualified lotteries are administratively feasible, make uncertainty explicit, can reduce reviewer burden and some forms of bias, and stop fine score differences from masquerading as objective knowledge.

At the frontier, a qualified lottery does not replace reason with chance. It replaces hidden chance with acknowledged chance. Reviewer assignment, panel composition, presentation order,

institutional familiarity, and small interpretive differences already introduce randomness. When evaluators cannot discriminate reliably among qualified proposals, a transparent random step may be more defensible than a ranking whose second decimal place is treated as destiny.

A lottery should not allocate every research euro. Different uncertainties require different selection mechanisms. Incremental work with mature methods can be judged through conventional excellence review. Some radical proposals need threshold-and-lottery funding. Others need a program manager willing to sponsor a coherent minority view. Validation infrastructure requires dedicated budgets because no individual proposal captures its full value. Long-horizon programs need reserves that allow renewal through multiple informative failures.

The percentages are not empirical optima. They are a design hypothesis intended to expose a more important principle: allocating the entire budget through one epistemic mechanism is an avoidable concentration of risk. A funder should be able to vary the shares by domain, evaluate the results, and publish how each lane performs.

The wildcard lane is especially useful against group convergence. UKRI's review of peer-review interventions describes variants known as the golden ticket, joker, or wildcard: a decision-maker may guarantee support for one proposal despite the ranking, subject to conflict-of-interest safeguards. The mechanism gives a reasoned minority a limited budget with which to challenge the consensus. Accountability comes from requiring the sponsor to state the thesis and the evidence that would later vindicate or defeat it.

The validation commons corrects a deeper market failure. Under the Discovery Flood, generation attracts prestige and private upside; replication, failed-branch memory, benchmark

maintenance, provenance, and epistemic or axiological review often do not. Yet these functions determine whether abundant claims become cumulative knowledge. They should not survive on whatever money remains after projects have promised their direct outputs.

The long-horizon reserve prevents smaller grants from becoming an ideology of instant results. Some ideas mature through years of negative evidence, redesigned instruments, and conceptual revision. The renewal rule should ask whether the program is improving the map of the problem, not whether it can announce a breakthrough every cycle. Patient capital becomes compatible with accountability when the evidence state shows why the next cycle is rational.

Longview Advisor can support allocation without becoming an autonomous funder. Its Funding Router should return a decision path rather than a universal score: conventional review, frontier lottery, micro-experiment, wildcard consideration, validation grant, long-horizon renewal, safety hold, or insufficient specification. Each route must cite the uncertainty structure that produced it and state what evidence would permit movement to another lane.

Funders should treat these mechanisms as experiments on their own institutions. They should preregister objectives, compare application and reviewer burden, analyze institutional and demographic concentration, track negative and positive outcomes over years, and publish failures. A funding agency that demands learning from scientists while refusing to learn from its allocation process reproduces the permission paradox internally.

The objective is not a perfectly fair machine for identifying genius. No such machine exists. The objective is a portfolio in which reputation is useful but not sovereign, chance is explicit where judgment becomes noisy, failure can buy information, and no single

committee can quietly determine which kinds of future are allowed to become testable.

Funding lane	Illustrative purpose and share
Conventional excellence	45% for work whose quality and feasibility can be discriminated reasonably well through established expertise and evidence.
Frontier qualified lottery	20% for anonymous, threshold-qualified, high-uncertainty proposals funded by random selection in small evidence tranches.
Wildcard or maverick	10% for proposals selected by accountable program managers or reviewers who defend a coherent minority judgment despite panel skepticism.
Validation commons	15% for replication, benchmarks, shared instruments, negative-result infrastructure, provenance, executable science, and independent assurance.
Long-horizon renewal reserve	10% for programs that repeatedly produce meaningful information but require years of iteration before their central objective can be settled.

CHAPTER 8

How the Strange Becomes Actionable

8.1 Positioning, Demonstration, and the First Dense Community

An innovator with an unfamiliar idea faces two temptations. The first is to explain the entire future until the audience understands the vision. The second is to hide the novelty and present the project as a minor improvement. The first often sounds grandiose; the second removes the reason the project matters. Good positioning builds a bridge without pretending the destination is nearby.

The strongest sequence begins with a painful present problem, proceeds to an observable mechanism, and only then opens the larger world. The problem gives the audience a reason to care. The mechanism gives it a reason to believe. The world gives it a reason to participate.

This is anchored novelty. Research on scientific impact suggests that influential work often combines a strongly conventional foundation with a smaller number of unusual combinations. Hargadon and Douglas described a related strategy in Edison's electric system: radical change became legible partly through design choices that connected the new system to familiar institutions and practices. The lesson is not to disguise the future dishonestly. It is to avoid requiring the audience to learn a new technology, category, vocabulary, business model, and social order at the same moment.

A demonstration is the most effective bridge. It compresses a future into an event that can be observed. For an Longview Advisor, the demonstration should not be a theatrical report on an easy

proposal. It should compare ordinary review with second-order review on a consequential case and reveal an implication, risk, or decisive experiment that qualified reviewers agree they would otherwise have missed. For executable science, the demonstration might take one contested claim and show a trace from source to evidence, code, review, replication, value assumptions, and an automatically generated next test.

The first market should be chosen by asymmetry of pain rather than enthusiasm for novelty. A research funder overwhelmed by AI-generated proposals, a venture team evaluating unfamiliar scientific infrastructure, or a regulatory unit facing technologies outside existing categories has a concrete reason to tolerate an imperfect new tool. Early users should also be able to recognize success. If the benefit is "better understanding of the future," the product will drift into consultancy. If the benefit is identifying decisive experiments, preventing a false rejection, or revealing a missing risk, the result can be audited.

Connectivity is the next criterion. Radical ideas spread through complex contagion: adoption often requires reinforcement from several credible peers. Five laboratories or funds that compare dossiers and share evaluation practices can create more momentum than five hundred isolated sign-ups. The objective is a dense local norm, not diffuse awareness.

This is why "go viral" is usually the wrong early strategy. Broad attention can freeze an immature category and attract audiences that reward spectacle rather than learning. The first community should contain users who can change the product, validators whose judgment matters to the next adopters, and builders of complementary infrastructure. Public scale comes after the mechanism and category have survived contact with this group.

The innovator should communicate at several resolutions. The shortest statement describes the changed capability. For the

Longview Advisor, it might be: "It reviews what follows if an idea works, not only whether the document looks credible." The next layer shows the evidence: a case in which the system altered a decision or designed a better test. The technical layer explains architecture, provenance, limitations, and evaluation. The strategic layer describes the institutional future the capability could enable. Mixing all layers into one pitch produces either vagueness or overload.

Positioning should also distinguish the role from adjacent categories. "AI reviewer" sounds like automated peer review and invites comparison on report quality. "Futurist agent" sounds speculative and overclaims prediction. "Longview Advisor" names the role rather than pretending to predict the future. Its practical descriptor - an AI agent for future-potential review - makes the function legible. The product is not selling foresight as charisma. It is selling structured attention to conditional consequences.

The same discipline applies to executable science. "Automating discovery" is too broad and activates both hype and fear. A stronger initial position is concrete: make claims auditable, experiments callable, and review traceable. The larger vision remains present, but it is earned through artifacts.

Community formation should avoid becoming mimetic self-protection. Innovators who feel misunderstood are vulnerable to creating circles in which criticism is interpreted as evidence of originality. A healthy frontier community shares benchmarks, failure logs, adversarial reviews, and replication tasks. Members support one another's right to test while remaining willing to kill weak mechanisms. Solidarity should protect exploration, not belief.

A useful organizational form is a frontier commons. Projects submit sources and receive interoperable implication dossiers. Evaluators can challenge branches, propose experiments, and declare value lenses. Validation organizations publish results. Funders can search

for high-option projects whose decisive tests are underfinanced. The commons does not certify truth. It lowers the cost of finding out.

Marketing then becomes part of epistemic design. A good message routes the right people toward the right evidence. It does not maximize approval. It reduces the distance between an unfamiliar mechanism and the community capable of testing it.

8.2 Capital That Buys Learning

Some investors and patrons deserve explicit praise because they perform a civilizational function that ordinary finance underprovides. They accept uncertainty before a category is mature, back people before milestones are conventional, and fund infrastructure whose value is distributed across a field. Their work is not charity toward eccentricity. It is the creation of information that other capital can later use.

The history of venture capital contains several versions of this role. Georges Doriot's American Research and Development Corporation helped establish a model for financing technical companies outside established industrial balance sheets. Arthur Rock became associated with the financing of Fairchild Semiconductor and Intel, helping convert scientific and engineering teams into companies that created new categories. Mike Markkula's early support of Apple combined capital, operating knowledge, and a belief that a young technical company could become more than a hobbyist business. The essential contribution was not prediction with certainty. It was the willingness to structure a bet before conventional institutions knew how to price it.

Modern frontier capital is more diverse. Some mechanisms fund the individual. Emergent Ventures has supported unusual projects through relatively open applications. Astera's residency provides time, salary, and research budgets for public-good work before it is

clear whether the result should become a company, nonprofit, standard, or open artifact. Foresight Institute's communities and funding programs cultivate technical work at the edge of AI, science, longevity, and related domains. Encode, developed by Pillar VC with ARIA in the United Kingdom, offers an AI-for-science pathway in which researchers can develop ideas before choosing between venture, open source, and other institutional forms. Fifty Years' 5050 program and Activate's fellowships help scientists cross the difficult transition from research to company creation without making the first support conditional on a polished venture persona.

Other mechanisms fund missing scientific infrastructure. Convergent Research pioneered Focused Research Organizations to build tools, datasets, and platforms that are too coordinated and engineering-heavy for a normal laboratory, yet too public-good-oriented for a conventional startup. Schmidt Sciences supports field-building programs and, in its trustworthy-AI agenda, emphasizes rigorous evaluation with construct and predictive validity. Renaissance Philanthropy develops funds and programs around scientific bottlenecks, AI for science, open-source infrastructure, autonomous instruments, and ambitious research-program design. ARIA uses program portfolios and opportunity seeds to pursue underexplored technological spaces with explicit tolerance for uncertainty.

Fast Grants, created during the COVID-19 emergency by Patrick Collison, Tyler Cowen, Patrick Hsu, and collaborators, demonstrated another important institutional possibility: scientific funding can be both rapid and serious. The model did not prove that all grantmaking should occur in days. It showed that many months of delay are not a law of nature and that flexible, fast capital can unlock research that conventional processes leave waiting.

Venture funds occupy a different point in the stack. Breakout Ventures focuses on science-driven companies and emphasizes the

ability to recognize inflection points before markets do. Fifty Years invests in companies aimed at large human and planetary needs. Lux Capital and DCVC have long backed deep-technology companies whose markets require scientific and technical conviction. Khosla Ventures explicitly describes seed investment as support for science or business experiments, including ideas with substantial probability of failure, provided the key technical risk and economic upside can be understood. These organizations are not interchangeable, and none is an automatic fit for a particular project. They are useful starting points because their mandates acknowledge category risk.

Business angels remain important because the first meaningful cheque often depends on a person rather than an institution. The best angel for a radical project is not necessarily the wealthiest or most famous. It is someone who can understand the mechanism, evaluate the founder's integrity under uncertainty, introduce credible validators, and tolerate milestones designed to learn rather than decorate a fundraising story. Technical founders, experienced operators, scientific philanthropists, and domain experts can provide this form of capital. Their reputation also creates the first bridge of social proof.

Innovators should evaluate investors as carefully as investors evaluate them. The relevant questions concern behavior under uncertainty. Has the investor backed a category before it had a name? Can the fund tolerate a public-good component or a long validation cycle? Does the partner understand the difference between a failed experiment and a failed company? Can the investor help recruit domain experts, access laboratories, navigate regulation, or build a community? Will the capital force premature value capture that prevents ecosystem adoption?

The project should be divided according to the economics of its components. In a program for executable science, open claim

standards, benchmark suites, reproducibility protocols, and neutral evaluation infrastructure have strong public-good characteristics. Philanthropy, public research funding, consortia, and focused research organizations may be appropriate. Enterprise review tools, orchestration platforms, provenance services, scientific copilots, and domain-specific validation products may support venture models. Public agencies can finance mission deployments and validation facilities. Universities and open-source communities can sustain shared knowledge layers.

A hybrid institutional structure may therefore be more honest than forcing everything into one company. A nonprofit or consortium can govern open standards. A company can commercialize robust implementations and services. A research program can test architectures that are too early for a product. Independent validation partners can prevent the commercial layer from grading its own claims. The capital architecture should mirror the epistemic architecture.

The approach to a frontier funder should not begin with "our idea is too advanced for ordinary investors." This statement supplies no evidence and makes resistance impossible to interpret. A stronger proposal identifies a bottleneck, explains why existing actors underproduce the missing capability, shows the mechanism, states what becomes possible if it works, defines a decisive milestone, and proposes an institutional form suited to the value created.

For the Longview Advisor, the most credible starting proposal is not a universal system for understanding the future. It is a domain-specific second-order reviewer with a benchmark and a live validation partner. The case might be AI-for-science research programs, scientific infrastructure investments, or technology proposals with measurable downstream implications. The first capital should finance evidence: historical temporal benchmarks,

prospective comparative trials, a structured dossier standard, and integration with one real decision workflow.

As of August 2026, the organizations named in this chapter provide a practical map rather than a promise. Programs, eligibility, and calls change. The durable method is to match the uncertainty to the institution. Person-first capital fits an exceptional builder with an unresolved form. Philanthropic capital fits public-good infrastructure and neglected validation. ARPA-like capital fits measurable, ambitious missions. Venture capital fits applications capable of capturing value at scale. Strategic partners fit access to laboratories, users, data, or distribution.

Capital is not only fuel. It is a permission technology and, at its best, a machine for purchasing information. The strongest frontier investors and funders do more than tolerate uncertainty. They organize it into a sequence of tests, finance the shared infrastructure that makes those tests credible, and allow the rest of society to arrive later with evidence rather than faith.

CHAPTER 9

Longview in the Executable Science Ecosystem

9.1 The Prospective Evaluation Layer

Longview is most useful when it is not built as an isolated AI product. Future-potential review depends on evidence retrieval, persistent project state, controlled agent execution, explicit provenance, correction, human governance, and increasingly machine-legible research artifacts. The broader Executable Science programme treats these capabilities as separate but composable layers. Longview occupies the prospective evaluation layer: it asks what a claim or technology could unlock and what evidence should be purchased next.

Longview Advisor belongs naturally in the Technology & Executable Science collection because it is not merely a thesis about institutions. It is a proposed piece of research infrastructure. The surrounding Axiologic technology books describe different layers of the same broader problem: how AI-assisted work can remain persistent, inspectable, evidence-bearing, bounded, and corrigible as models become more capable. Longview adds a missing layer of prospective judgment.

AssistOS is the closest operational substrate. Its premise is that AI work needs an operating layer in which models, agents, artifacts, memory, permissions, and human collaborators can persist beyond a single chat. Longview needs exactly such a substrate because a future-potential review is not a one-shot response. The dossier must live beside the proposal, evidence, experiments, reviewer annotations, and later results. AssistOS can therefore host

Longview as a specialised agentic capability rather than Longview having to invent its own workspace.

MRP-VM contributes the harness discipline. A high-stakes adviser should have pinned sources, permitted tools, model versions, policies, output contracts, qualification tests, and rollback targets. Longview's trustworthiness depends less on one foundation model than on this surrounding harness. A model can be replaced without silently changing the semantic contract of the evaluation.

RAG and Evidence Portfolio Retrieval contribute the evidence layer. Longview cannot assess future potential from the proposal alone. It needs related work, failed attempts, historical precedents, enabling technologies, contradicting evidence, and alternative mechanisms. Ordinary relevance retrieval tends to return many sources saying similar things. An evidence portfolio instead tries to assemble the configuration needed to test a claim. That is a better match to review.

Executable Natural Language and Executable Scientific Intelligence contribute the semantic and computational layer. When a decisive claim can be made explicit, it should become more than prose. A source-grounded controlled representation can connect the claim to its assumptions, variables, evidence, and executable checks. Dynamic SOP Lang circuits can express bounded analyses without pretending that every judgment can be formalised. Longview can remain linguistically flexible at the frontier while progressively compiling stable pieces into inspectable operations.

The Frontier Is Correction contributes the normative test for the whole stack: a scientific system matters if reality can force revision. Longview therefore must not be measured by how persuasive its initial dossier appears. It must be measured by how appropriately the dossier changes when experiments fail, evidence contradicts a branch, or a human expert identifies a mistaken representation. The Future of Research Infrastructure provides the persistent

project state needed for this revision. Coherence Pressure provides a warning: if the adviser resolves every tension into one elegant story, it has failed even when its prose is excellent.

Taken together, these components suggest an ecosystem rather than a monolith. Longview can be implemented first with existing LLMs and ordinary databases. It can then progressively adopt evidence portfolios, controlled semantic representations, agentic harnesses, dynamic circuits, and living research state where each addition earns its complexity by improving a benchmark. This incremental path is important for a Horizon Europe research project because it creates clear technical hypotheses and integration milestones rather than bundling every speculative technology into one unverifiable promise.

Table 3 · Longview and adjacent Axiologic technology concepts

Technology or research concept		Relationship to Longview Advisor
AssistOS		Possible operating and collaboration layer for persistent projects, agents, artifacts, permissions, and human review.
MRP-VM		Custom agentic harness discipline: pinned sources, tools, policies, qualification gates, traces, and rollback for repeatable Longview capabilities.
RAG and Evidence Portfolio Retrieval		Retrieval of balanced support, contradiction, lineage, alternatives, and negative evidence rather than only nearest semantic passages.
Executable Natural Language / SLEnglish		Candidate controlled semantic layer for source-grounded claims that need deterministic parsing, comparison, validation, or execution.
Executable Scientific Intelligence / SOP Lang Circuits		Bounded task-local computations and dynamic circuits for formalizable parts of review, evidence checks, and experiment execution.

Technology or research concept	Relationship to Longview Advisor
The Frontier Is Correction	Correction principle: a Longview judgment must change when evidence, replication, or criticism changes the underlying state.
The Future of Research Infrastructure	Living project state in which claims, evidence, failures, decisions, and authority remain connected over time.
Coherence Pressure	Failure model reminding Longview not to smooth disagreement or novelty into a single persuasive explanation.

9.2 A Research Programme and European Research Use Case

The European context is unusually favourable to testing this idea, although the book does not claim automatic eligibility for any call. The European Commission's AI in Science Strategy treats AI as a technology that can reshape scientific work from literature review to laboratory automation and has created RAISE, the Resource for AI Science in Europe, as a virtual institute pooling compute, data, talent, and funding. The 2026-2027 Horizon Europe work programme includes a horizontal AI in Science call. The 2027 call HORIZON-RAISE-2027-01 is scheduled to open on 22 September 2026 with a deadline of 2 February 2027; its Scientific Discovery topic has an indicative budget of EUR 29 million and expects projects around EUR 10 million. Longview's relevance is not that it alone answers that topic, but that AI-generated scientific discovery creates exactly the validation and prioritisation problem the technology is designed to study.

A credible Horizon use case would place Longview inside an AI-for-science consortium rather than present it as a universal grant-grading machine. Automated discovery agents would

generate hypotheses or experimental directions. Longview would provide an independent prospective-evaluation layer that reconstructs each candidate, searches evidence and prior art, explores downstream scientific and societal implications, selects high-information experiments, and tracks how the assessment changes after laboratory results. Human scientists and programme stakeholders would compare Longview-assisted prioritisation against existing expert selection. This creates a measurable research contribution to the validation bottleneck.

A second Horizon use case is research infrastructure. European Research Infrastructures explicitly include knowledge facilities, scientific data infrastructures, computing systems, and communication networks. Longview can be studied as a digital service over a living research state, enabling cross-project evidence tracking, impact exploration, and event-triggered review. The INFRATECH destination's emphasis on next-generation scientific instrumentation, tools, methods, and advanced digital solutions illustrates the institutional category into which such work can fit, even when a particular annual call is already closed.

A third line is trustworthy agentic AI. The Cluster 4 work programme has funded research on next-generation AI agents, autonomy, robustness, planning, memory, and reasoning. Longview is a high-stakes agentic use case with unusually clear requirements for bounded authority, privacy, provenance, and human override. It can therefore serve as a demanding benchmark for agent architectures developed in broader trustworthy-AI projects.

A fourth use case is the grant system itself, but this should begin as research, not deployment by stealth. A funding body could provide de-identified historical proposals and outcomes under secure conditions. Longview would run temporal evaluations using only information available at the time of the original decision. Researchers would test whether it identifies eventual high-impact

projects earlier, whether it over-rewards fashionable narratives, whether reputation masking changes results, and whether its suggested experiments would have reduced uncertainty. A prospective pilot could then compare ordinary review with reviewer-plus-Longview review on a small, authorised funding programme. The decision would remain human.

The research hypothesis is precise enough to fail: reviewer access to a structured Future-Potential Review will increase detection of consequential, testable implications and improve the selection of information-gaining experiments without increasing false-positive enthusiasm, confidence inflation, or systematic bias. Secondary hypotheses concern reduced dependence on reputation, improved calibration, shorter review time for complex proposals, better retention of productive failures, and improved portfolio diversity.

Evaluation should not rely on retrospective hero stories. Historical benchmarks need successes, failures, harms, incremental projects, and ideas that were unusual but unimportant. Temporal cutoffs prevent the system from using knowledge that appeared later. Structured perturbations test whether it reacts correctly when mechanism, timeline, evidence, identity, or rhetorical framing changes. Prospective trials measure how human reviewers change their decisions after seeing Longview and whether those changes improve eventual outcomes. The system should be considered harmful if it mainly makes reviewers more confident.

This research programme also aligns with changes already visible in European funding. The 2026-2027 work programme expands lump-sum funding, two-stage calls, shorter first-stage proposals, and blind evaluation for many topics. The EIC Advanced Innovation Challenges pilot uses a EUR 300,000 lump sum for up to nine months before a possible larger second stage. These mechanisms do not prove Longview's design, but they show that Europe is already

experimenting with the premise that high-risk innovation can be funded as staged learning rather than as a single long prediction.

The practical proposal is therefore modest in form and ambitious in consequence. Build Longview as a research-grade adviser inside one or two real decision workflows. Compare an LLM-heavy architecture with a progressively neuro-symbolic one. Integrate with evidence retrieval and persistent research state. Run blinded and prospective evaluations. Publish the failure cases. Only then ask whether the technology deserves a broader role in European research administration, philanthropy, venture science, or corporate R&D.

Table 4 · Illustrative European research positioning as of August 2026

The first Longview prototype should be deliberately narrow. It accepts English-language research and innovation proposals in one or two domains with strong literature and measurable validation pathways. The system returns a versioned Longview Dossier and never makes an autonomous funding decision. The primary benchmark is whether expert reviewers obtain useful, non-obvious, source-grounded implications and better next-test choices than they obtain from ordinary review or a general-purpose chatbot.

Inputs consist of a source package, a decision context, a lens profile, identity metadata stored separately for the Reputation Firewall, and optional programme rules. The source package is immutable and versioned. Retrieval occurs through authorised scholarly and web sources with timestamps. Every generated claim is labelled as source reconstruction, retrieved evidence, analogy, system inference, scenario assumption, value judgment, or human annotation.

The LLM-heavy prototype should implement secure intake, source reconstruction, evidence portfolio retrieval, novelty mapping, independent future-potential generation, counterfactual criticism,

stakeholder and value analysis, experiment design, dossier rendering, and audit logging. Model diversity is desirable but not assumed to create independence. Independence is measured through disagreement patterns, source diversity, and perturbation tests.

The neuro-symbolic prototype adds typed objects for claims, evidence, assumptions, implications, stakeholders, values, experiments, and decisions. It should support provenance, contradictions, versioning, alternative interpretations, and selective recomputation. Candidate controlled-language or SOP Lang representations are introduced only where they improve fidelity or execution. The project should reject symbolic structure that merely makes uncertain judgments look precise.

The historical benchmark must be temporal. Cases are frozen at a point before the later outcome is known. The set includes eventual high-impact successes, ordinary successes, failures, harmful ideas, incremental projects, and unusual projects that remained unimportant. Reviewers and Longview receive only information available at the historical cutoff. Outcome measures include implication coverage, calibration, experiment quality, false-positive enthusiasm, sensitivity to prestige, and hindsight leakage.

The perturbation benchmark changes one property at a time while preserving the rest of the proposal. Identity, institutional prestige, rhetoric, citations, timelines, baselines, mechanism details, impact claims, and hidden contradictions can be manipulated. Longview should change when causal or evidentiary content changes and remain relatively stable when irrelevant prestige or stylistic polish changes.

The prospective benchmark is a field study with an authorised funding, investment, or R&D partner. Reviewers are randomised to ordinary review or ordinary review plus Longview. The study measures decision time, confidence, proposal ranking, diversity of

selected experiments, reviewer agreement, novelty detection, usefulness of critiques, and later evidence from funded projects. A crucial harm metric is confidence inflation: the technology fails if it increases certainty without improving decision quality.

Privacy and governance are first-class requirements. Confidential proposals remain inside approved infrastructure. No model receives more data than its role requires. External services are disabled when policy or confidentiality requires local processing. The system records model versions, prompts or policies, retrieved sources, human interventions, and final overrides. Users can appeal a Longview inference by challenging its source, semantic reconstruction, implication chain, or value lens.

The research programme should remain open to a negative result. Longview may prove useful only for certain domains, only for experiment design rather than impact analysis, or only as reviewer feedback rather than proposal triage. A neuro-symbolic layer may fail to justify its engineering cost. A reputation firewall may reduce one bias while removing legitimate execution information. The purpose of the project is to find these boundaries before institutional deployment.

European direction	Possible Longview role
HORIZON-RAISE-2027-01-01 Scientific Discovery	A validation and prioritisation layer inside an AI-for-science consortium: future-potential review of generated hypotheses, experiment selection, and prospective benchmarking. This is an illustrative fit, not an eligibility claim.
European AI in Science Strategy / RAISE	Decision-support infrastructure for the validation bottleneck created when AI increases the rate of scientific conjecture and experimentation.
Research Infrastructures / INFRATECH	Advanced digital service for persistent evidence, cross-project review, impact

European direction	Possible Longview role
Cluster 4 trustworthy next-generation agents	exploration, and machine-assisted research governance. High-stakes agentic benchmark requiring bounded authority, memory, provenance, privacy, adversarial evaluation, and human override.
EIC staged innovation funding	Longview as a tool for selecting the next information-gaining milestone and for comparing projects between short validation and larger continuation stages.

EPILOGUE

Do Not Let the Future Die in Triage

The permission paradox once described a recurring inconvenience of progress. In the age of AI, it may become a central political and epistemic problem because it now sits inside a Discovery Flood.

The production of candidate novelty is beginning to accelerate beyond the reading, validation, and adoption capacity of existing institutions. The predictable response is more compression, more ranking, more dependence on reputation, and more automated review. These mechanisms are necessary. Left uncorrected, they become flood barriers that protect institutional throughput by discarding whatever cannot be processed quickly. They will systematically favor ideas that resemble what institutions already know how to approve.

The future may therefore be lost in a surprisingly ordinary place: the triage layer. Not because a government prohibits imagination, but because a system summarizes an unfamiliar proposal into a familiar category, assigns it a moderate score, and moves on. Not because reviewers are malicious, but because they are overloaded. Not because society deliberately chooses stagnation or dystopia, but because public-interest experiments wait while concentrated actors with their own capital and infrastructure proceed. Institutional inertia is not neutral when the opportunity cost is a foregone capability that could have benefited millions.

The answer is not universal permission and not the romantic abolition of gatekeeping. It is to replace gatekeeping as a bottleneck with evaluation as an active learning system. We need second-order review that examines what follows if an idea works, validation

infrastructure that turns uncertainty into tests, smaller evidence tranches with long institutional patience, reputation firewalls, qualified lotteries where rankings are not trustworthy, and explicit accounting for the costs of delay. We need more independent searches, more diverse portfolios, more reversible pilots, and public memory for productive failure.

Humanity's mimetic capacities will not disappear. Nor should they. Copying is how successful discoveries spread. The challenge is to prevent imitation from deciding the search before reality has had a vote.

The AI revolution may give us a rare opportunity to enlarge both the frontier of the possible and the machinery for inspecting it. The Discovery Flood is not a reason to retreat. It is a reason to build faster epistemic infrastructure. The opportunity will be wasted if machines generate a million possible futures and institutions learn only to summarize them.

A civilization worthy of its tools must learn to absorb novelty without confusing speed of triage with quality of judgment.

It must ask what good may become possible tomorrow, what evidence can be created now, and who has the right to prevent us from finding out.

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